

Lab Tasks

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Section: D

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Course: Programming Fundamentals - 2035

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Program: (BS) - Computer Science

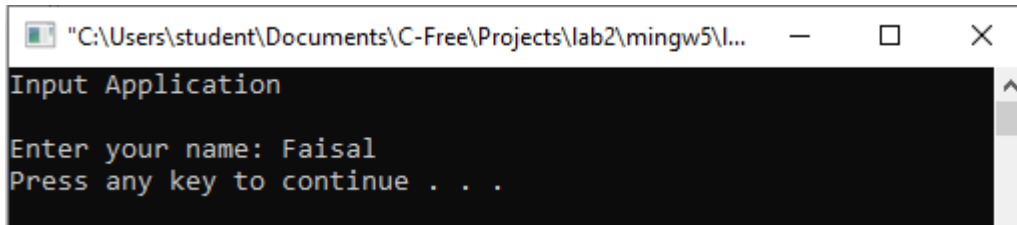
Session: Fall 2022

Task 1: Write a C++ program to take input from the user and display it on the screen.

Code

```
#include <iostream.h>
using namespace std;
int main()
{
    cout<<"Input Application \n";
    char name;
    cout<<"\nEnter your name: ";
    cin>>name;
}
```

Output

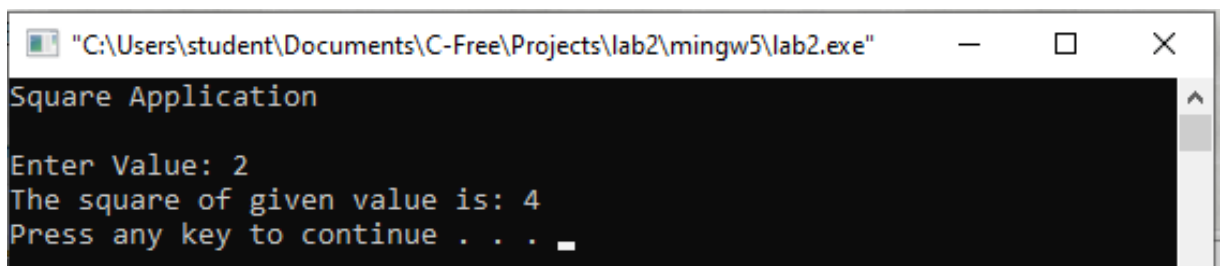


Task 2: Write a C++ program to take input from user and calculate its square-root.

Code

```
#include <iostream.h>
using namespace std;
int main()
{
    cout<<"Square Application \n";
    int a,square;
    cout<<"\nEnter Value: ";
    cin>>a;
    square=a*a;
    cout<<"The square of given value is: "<<square<<endl;
}
```

Output



Task 3 Write a C++ program that asks the user to enter two numbers, obtains the two numbers from the user, and prints the sum, product, difference, and quotient of the two numbers.

Code

```
#include <iostream.h>
using namespace std;
int main()
{
    cout<<"Calculator Application\n";
    int a,b,sum,sub,multiply,divide;
    cout<<"\nEnter first Value\n";
    cin>>a;
    cout<<"Enter socond value\n";
    cin>>b;
```

```

sum=a+b;
sub=a-b;
multiply=a*b;
divide=a/b;

cout<<"\n Result of addition is: "<<sum;
cout<<"\n Result of subtraction is: "<<sub;
cout<<"\n Result of multiplication is: "<<multiply;
cout<<"\n Result of division is: "<<divide<<endl;
return 0;
}

```

Output

```

Calculator Application

Enter first Value
50
Enter socond value
25

Result of addition is: 75
Result of subtraction is: 25
Result of multiplication is: 1250
Result of division is: 2
Press any key to continue . . .

```

Task 4: Write a program to take two numbers as input from user and return swap results.

Code

```

#include <iostream>
using namespace std;
int main()
{
    cout << "Swap Results Application\n";
    int a, b, temporary;
    cout << "\nEnter value a\n";
    cin >> a;
    cout << "Enter value b\n";
    cin >> b;
    temporary = a;
    a = b;
    b = temporary;
    cout << "After Swapping" << endl;
    cout << "a=" << a << ", b=" << b << endl;
    return 0;
}

```

Output

```

Microsoft Visual Studio Debug Console

Swap Results Application

Enter value a
20
Enter value b
10
After Swapping
a=10, b=20

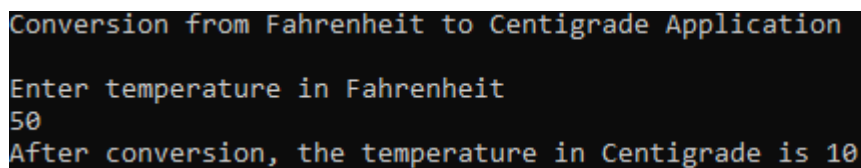
```

Task 5: Write a C++ program that takes the temperature in Fahrenheit as an input argument and returns the centigrade temperature.

Code

```
#include <iostream>
using namespace std;
int main()
{
    cout<<"Conversion from Fahrenheit to Centigrade Application\n";
    float c, f;
    cout << "\nEnter temperature in Fahrenheit \n";
    cin >> f;
    c = 5 * (f - 32) / 9;
    cout <<"After conversion, the temperature in Centigrade is "<<c<<endl;
    return 0;
}
```

Output

A screenshot of a terminal window showing the output of the C++ program. The text is as follows:
Conversion from Fahrenheit to Centigrade Application

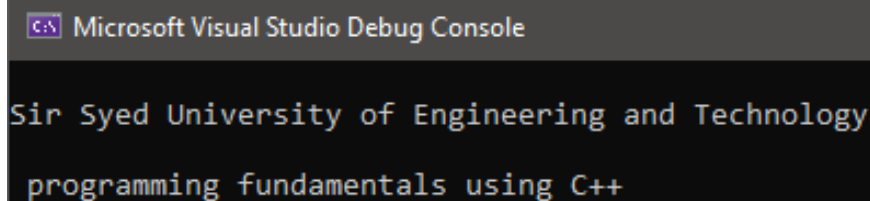
Enter temperature in Fahrenheit
50
After conversion, the temperature in Centigrade is 10

Lab Session 1

Task 1: What is the output from the following C++ code fragment?

```
#include <iostream>
using namespace std;
int main(void)
{
    cout<<"\nSir Syed University of Engineering and Technology"<<endl;
    cout<<"\t \n programming fundamentals using C++" <<endl;
    return 0;
}
```

Output

A screenshot of the Microsoft Visual Studio Debug Console showing the output of the C++ program. The text is as follows:
Sir Syed University of Engineering and Technology

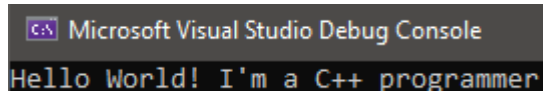
programming fundamentals using C++

Lab Session 2 (In-Lab Tasks)

Task #1: What will be the output of the following C++ code fragment?

```
#include <iostream>
Using namespace std;
int main ()
{
    cout << "Hello World! "; // prints Hello World!
    cout << "I'm a C++ programmer"; // prints I'm a C++ program
    return 0;
}
```

Output

A screenshot of the Microsoft Visual Studio Debug Console showing the output of the C++ program. The text is as follows:
Hello World! I'm a C++ programmer

Task #2: Find out errors (if any) in the following C++ program

```
1. #include<iostream>
2. using namespace std;
3. void main()
4. { /* This is the main function
5.     cout<<'C++ Programming questions and answers';
6. }
```

Errors

- (line 3) return type of 'main' should be 'int' instead of 'void'
- (line 4) comment unclosed
- (line5) double quotes should be used instead of single quotes to print a statement

Task #3: Write down the output of the following code in the box

```
#include <iostream>
using namespace std;
int main()
{
int num1, num2;
int sale;
char first;
num1 = 4; cout << "num1 = " << num1 << endl;
num2 = 4 * 5 - 11;
cout << "num2 = " << num2 << endl;
sale = 2 * 1000;
cout << "sale = " << sale << endl;
return 0;
}
```

Output

```
num1 = 4
num2 = 9
sale = 2000
```

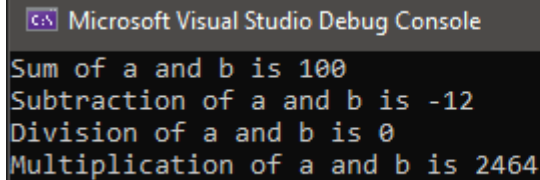
Lab Session 2 (Post-Lab Tasks)

Task 1: Write C++ code, in which initialize two variable a and b with 44 and 56. Print addition, subtraction, division and multiplication of a with b.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int sum, multiply, sub, divide;
    int a = 44, b = 56;
    sum = a + b;
    sub = a - b;
    divide = a / b;
    multiply = a * b;
    cout << "Sum of a and b is " << sum << endl;
    cout << "Subtraction of a and b is " << sub << endl;
    cout << "Division of a and b is " << divide << endl;
    cout << "Multiplication of a and b is " << multiply << endl;
    return 0;
}
```

Output



```
C:\> Microsoft Visual Studio Debug Console
Sum of a and b is 100
Subtraction of a and b is -12
Division of a and b is 0
Multiplication of a and b is 2464
```

Task 2: Write down C++ code of following. Create two variables T and U. Initialize variables with 5 and 2. Print square of T and U (square is simple multiplication of T*T or U*U).

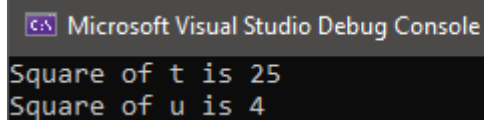
Code

```
#include <iostream>
using namespace std;
int main()
{
    int square1, square2;
    int t = 5, u = 2;

    square1 = t * t;
    square2 = u * u;

    cout << "Square of t is " << square1 << endl;
    cout << "Square of u is " << square2 << endl;
    return 0;
}
```

Output



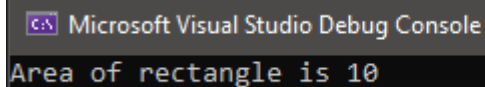
```
C:\> Microsoft Visual Studio Debug Console
Square of t is 25
Square of u is 4
```

Task 3: Write a C++ program that calculates the area of a rectangle as per the formula shown (Area = base x height). The values of base and height are 5 and 8 respectively.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int area;
    int b = 5, h = 2;
    area = b * h;
    cout << "Area of rectangle is " << area << endl;
    return 0;
}
```

Output



```
C:\> Microsoft Visual Studio Debug Console
Area of rectangle is 10
```

Task 4: Write C++ code, in which the length & breadth of a rectangle are 13 and 14. Also, the radius of a circle is 10. Write a program to calculate the area & perimeter of the rectangle and the area & circumference of the circle.

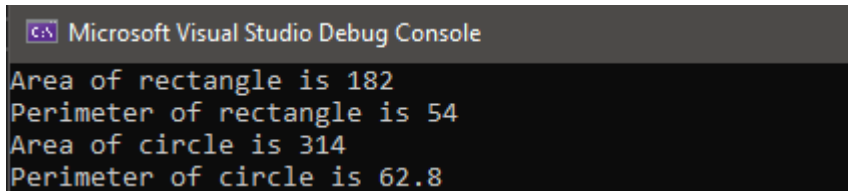
Code

```
int main()
{
    int area_rect, perimeter_rect, area_circle;
    float perimeter_circle, y=3.14;
    int l = 13, b = 14, r=10, x=2;

    area_rect = l * b;
    perimeter_rect = (x * l) + (x * b);
    area_circle = y * (r * r);
    perimeter_circle = x * y * r;

    cout << "Area of rectangle is " << area_rect << endl;
    cout << "Perimeter of rectangle is " << perimeter_rect << endl;
    cout << "Area of circle is " << area_circle << endl;
    cout << "Perimeter of circle is " << perimeter_circle << endl;
    return 0;
}
```

Output

The screenshot shows the Microsoft Visual Studio Debug Console with the following output:

```
Area of rectangle is 182
Perimeter of rectangle is 54
Area of circle is 314
Perimeter of circle is 62.8
```

Task 5: Write a C++ program that takes obtained marks of a student in five different subjects, each subject has maximum 100 numbers. Calculate total numbers (Obtain numbers out of 500) and percentage of student and print it on the output screen.

Code

```
#include <iostream>
using namespace std;
int main()
{
    float sub1, sub2, sub3, sub4, sub5, total, per;

    cout << "Enter marks obtained in subject 1 " << endl;
    cin >> sub1;
    cout << "Enter marks obtained in subject 2 " << endl;
    cin >> sub2;
    cout << "Enter marks obtained in subject 3 " << endl;
    cin >> sub3;
    cout << "Enter marks obtained in subject 4 " << endl;
    cin >> sub4;
    cout << "Enter marks obtained in subject 5 " << endl;
    cin >> sub5;

    total = sub1 + sub2 + sub3 + sub4 + sub5;
    per = (total / 500) * 100;

    cout << "Total marks obtained in all 5 subjects " << total << endl;
    cout << "Percentage of all 5 subjects " << per << endl;
    return 0;
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter marks obtained in subject 1
70
Enter marks obtained in subject 2
75
Enter marks obtained in subject 3
90
Enter marks obtained in subject 4
75.5
Enter marks obtained in subject 5
90.25
Total marks obtained in all 5 subjects 400.75
Percentage of all 5 subjects 80.15
```

Task 6: Write down output of following code in box

Code

```
#include <iostream>
using namespace std;
int main()
{
    int a = 3, b = 4;
    cout << "a= " << a << " b= " << b << endl;
    a = 6;
    cout << "a= " << a << " b= " << b << endl;
    b = 9;
    cout << "a= " << a << " b= " << b << endl;
    a = 0;
    b = 0;
    cout << "b= " << b << " b= " << b << endl;
    return 0;
}
```

Output

```
a= 3 b= 4
a= 6 b= 4
a= 6 b= 9
b= 0 b= 0
```

Task 7: Find out errors in the following C++ programs:

```
a) 1. #include <iostream>
    2. using namespace std;
    3. int (main)
    4. {
    5.     cout << "Hello, World!";
    6.     cout << 'Sir Syed University of Engineering and Technology';
    7.     cout << "COMPUTER SCIENCE",
    8.     Cout << "PROGRAMMING FUNDAMENTALS";
    9.     return 0;
    10. }
```

Errors

- (line 3) 'main' cannot be used as global variable name
- (line 6) double quotes should be used instead of single quotes to print a statement
- (line 7) semicolon ; should be used instead of coma , to print a statement
- (line 8) c of cout should be in small letter


```
b) 1. #include [iostream]
    2. int main()
    3. {
    4.     std::cout << "Hello,World!".
    5.     return 0;
    6. }
```

Errors

- (line 1) < > should be used instead of []
- (line 4) semicolon ; should be used instead of full stop to print a statement

Lab Session 3 (In-Lab Tasks)

Task 1: Write a program that takes four floating numbers from the keyboard and prints their sum, product, and average.

Code

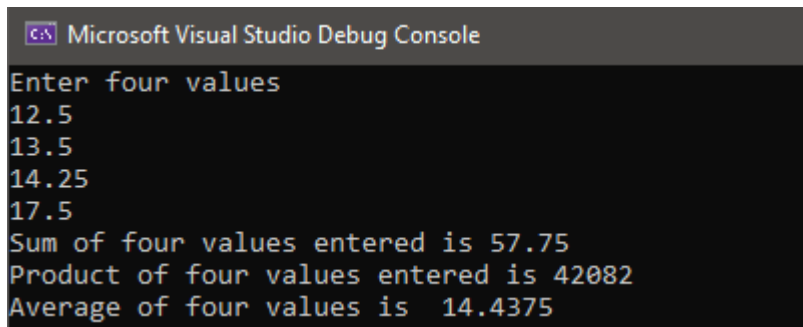
```
#include <iostream>
using namespace std;
int main()
{
    float sum, product, average;
    float a, b, c, d;

    cout << "Enter four values" << endl;
    cin >> a >> b >> c >> d;

    sum = a + b + c + d;
    product = a * b * c * d;
    average = sum / 4;

    cout << "Sum of four values entered is " << sum << endl;
    cout << "Product of four values entered is " << product << endl;
    cout << "Average of four values is " << average << endl;
    return 0;
}
```

Output



```
Microsoft Visual Studio Debug Console
Enter four values
12.5
13.5
14.25
17.5
Sum of four values entered is 57.75
Product of four values entered is 42082
Average of four values is 14.4375
```

Task 2: Write a program to calculate the area of the circle, taking the value of the radius from the user.

Code

```
#include <iostream>
using namespace std;
int main()
{
    float area_circle, r;
    cout << "Enter value of radius\n";
    cin >> r;
```

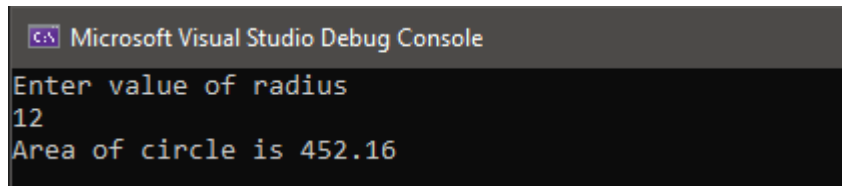
```

    area_circle = 3.14 * (r * r);

    cout << "Area of circle is " << area_circle << endl;
    return 0;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter value of radius
12
Area of circle is 452.16

```

Task#3: Write a program that prints to calculate the age in days by using the formula:
 $\text{days} = \text{years} * 365$.

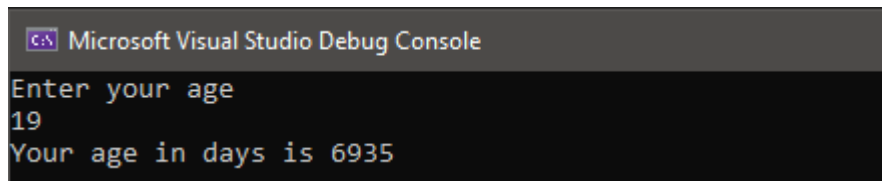
Code

```

#include <iostream>
using namespace std;
int main()
{
    int days, years;
    cout << "Enter your age\n";
    cin >> years;
    days = years * 365;
    cout << "Your age in days is " << days << endl;
    return 0;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter your age
19
Your age in days is 6935

```

Task 4: Write a C++ program to convert temperature from degree Celsius to Fahrenheit.

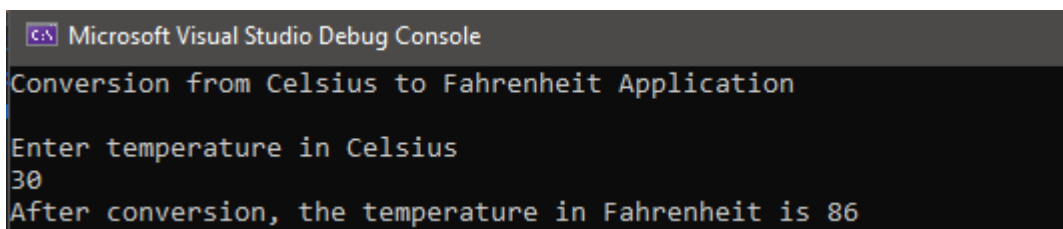
Code

```

#include <iostream>
using namespace std;
int main()
{
    cout << "Conversion from Celsius to Fahrenheit Application\n";
    float c, f ;
    cout << "\nEnter temperature in Celsius \n";
    cin >> c;
    f = (c * 9 / 5) + 32;
    cout << "After conversion, the temperature in Fahrenheit is " << f;
    return 0;
}

```

Output



```

Microsoft Visual Studio Debug Console
Conversion from Celsius to Fahrenheit Application
Enter temperature in Celsius
30
After conversion, the temperature in Fahrenheit is 86

```

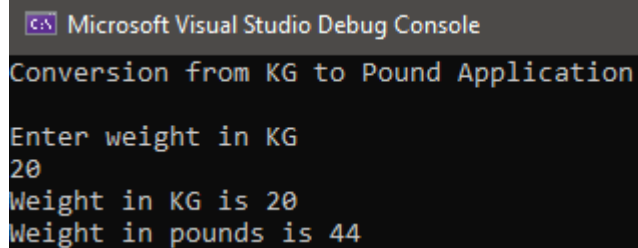
Lab Session 3 (Post-Lab Tasks)

Task 1: Write a C++ program that takes a person's weight in KG, and converts it into pounds. Your program should output both weights. (Note that 1 KG is 2.2 pounds.)

Code

```
#include <iostream>
using namespace std;
int main()
{
    cout << "Conversion from KG to Pound Application\n";
    float kg, p ;
    cout << "\nEnter weight in KG \n";
    cin >> kg;
    p = 2.2 * kg;
    cout << "Weight in KG is " << kg << endl;
    cout << "Weight in pounds is " << p;
    return 0;
}
```

Output



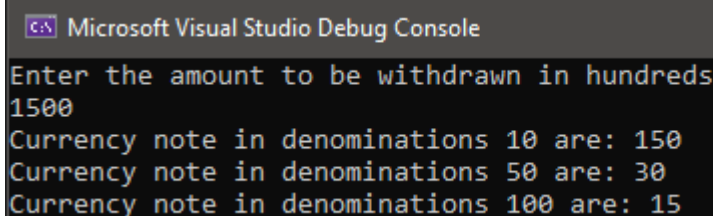
```
Microsoft Visual Studio Debug Console
Conversion from KG to Pound Application
Enter weight in KG
20
Weight in KG is 20
Weight in pounds is 44
```

Task 2: A cashier has currency notes of denominations 10, 50 and 100. If the amount to be withdrawn is input through the keyboard in hundreds, find the total number of currency notes of each denomination the cashier will have to give to the withdrawer. Write a C++ program to implement the above scenario.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int amount, t, f, h;
    cout << "Enter the amount to be withdrawn in hundreds\n";
    cin >> amount;
    h = amount / 100;
    f = amount / 50;
    t = amount / 10;
    cout << "Currency note in denominations 10 are: " << t << endl;
    cout << "Currency note in denominations 50 are: " << f << endl;
    cout << "Currency note in denominations 100 are: " << h << endl;
    return 0;
}
```

Output



```
Microsoft Visual Studio Debug Console
Enter the amount to be withdrawn in hundreds
1500
Currency note in denominations 10 are: 150
Currency note in denominations 50 are: 30
Currency note in denominations 100 are: 15
```

Task 3: Write a C++ program that calculates the monthly salary of an employee. The salary is calculated after the deduction of the following taxes. Income Tax: 10%. Global Life Insurance: 2.75%. Pension Plan: 2.3%. Health Insurance: 200.0 rupees. Your program should prompt the user to input the gross amount and your program should show each tax and net salary (after tax deduction). Your output should be in the same format (not the same values) as shown in the example below.

Code

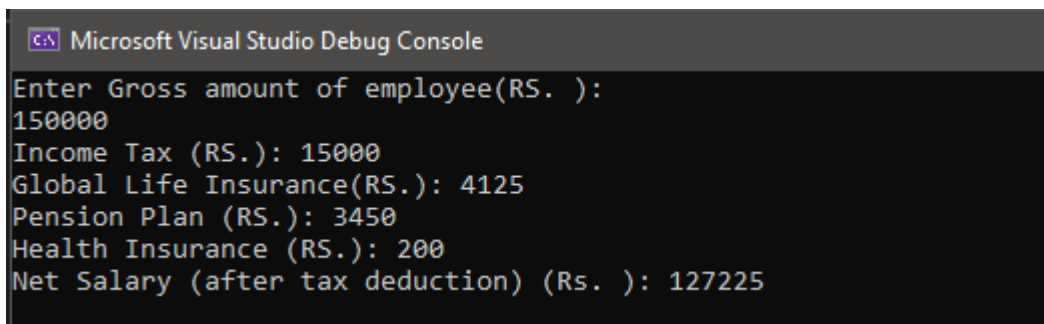
```
#include <iostream>
using namespace std;
int main()
{
    int salary, total, reduction;
    float i, l, p, h;
    cout << "Enter Gross amount of employee(RS. ):\n";
    cin >> salary;

    i = (10.0 / 100) * salary;
    l = (2.75 / 100) * salary;
    p = (2.3 / 100) * salary;
    h = 200.0;
    total = i + l + p + h;
    reduction = salary - total;

    cout << "Income Tax (RS.): " << i << endl;
    cout << "Global Life Insurance(RS.): " << l << endl;
    cout << "Pension Plan (RS.): " << p << endl;
    cout << "Health Insurance (RS.): " << h << endl;
    cout << "Net Salary (after tax deduction) (Rs. ): " << reduction << endl;

    return 0;
}
```

Output

A screenshot of the Microsoft Visual Studio Debug Console. The title bar at the top reads "C:\ Microsoft Visual Studio Debug Console". The console text shows the program's execution: it prompts for the gross amount, receives 150000, and then displays the calculated taxes and net salary. The output is as follows:

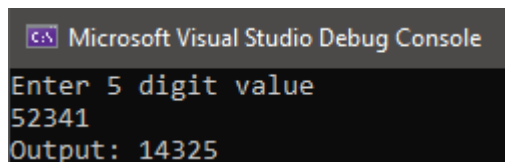
```
Enter Gross amount of employee(RS. ):
150000
Income Tax (RS.): 15000
Global Life Insurance(RS.): 4125
Pension Plan (RS.): 3450
Health Insurance (RS.): 200
Net Salary (after tax deduction) (Rs. ): 127225
```

Task 4: Write a C++ program that takes a 5-digit number in variable X. Your program should store the reverse of entered number into variable Y. Your program should also print it on screen.

Code

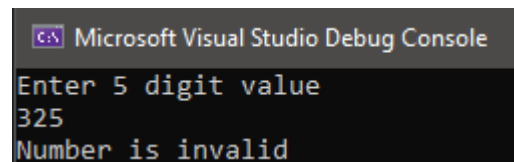
```
#include <iostream>
using namespace std;
int main()
{
    int num, reverse, d1, d2, d3, d4, d5;
    cout << "Enter 5 digit value\n";
    cin >> num;
    if (num >= 10000 && num <= 99999) {
        d5 = num % 10;
        num = num / 10;
        d4 = num % 10;
        num = num / 10;
        d3 = num % 10;
        num = num / 10;
        d2 = num % 10;
        d1 = num / 10;
        reverse = d5 * 10000 + d4 * 1000 + d3 * 100 + d2 * 10 + d1;
        cout << "Output: " << reverse;
    }
    else {
        cout << "Number is invalid";
    }
    return 0;
}
```

Output



Microsoft Visual Studio Debug Console

```
Enter 5 digit value
52341
Output: 14325
```



Microsoft Visual Studio Debug Console

```
Enter 5 digit value
325
Number is invalid
```

Lab Session 4 (In-Lab Tasks)

Task: 1 Mention the output for the following program

```
1. #include<stdio.h>
2. void main()
3. {
4.     int a = 100;
5.     if (a > 10)
6.         cout << "Shahid Afridi";
7.     else if (a > 20)
8.         cout << "Shoaib Akhtar";
9.     else if (a > 30)
10.        cout << "Kamran Akmal";
11. }
```

Output

There are several **errors** in this program therefore the program will not run and there will be no output

- (line 1) <iostream> should be used instead of <stdio.h>. #include<iostream> tells the processor to include the contents of the file <iostream>
- (line 2) return type of 'main' should be 'int' instead of 'void' because the void main() indicates that the main() function will not return any value, but the int main() indicates that the main() can return integer type data.
- (line 6, 8, 10) std:: is missing before cout. We need std:: to specify what "namespace" cout belongs to.

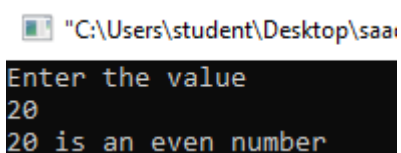
Task 2: Write a program that takes a number as input from the user and checks whether the number is even or odd Using if-else.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int i;
    cout << "Enter the value\n";
    cin >> i;

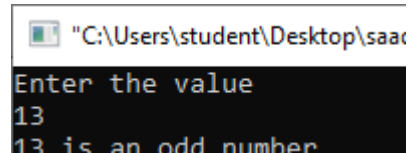
    if (i % 2 == 1) {
        cout << i << " is an odd number" << endl;
    }
    else {
        cout << i << " is an even number" << endl;
    }
}
```

Output



"C:\Users\student\Desktop\saar"

```
Enter the value
20
20 is an even number
```



"C:\Users\student\Desktop\saar"

```
Enter the value
13
13 is an odd number
```

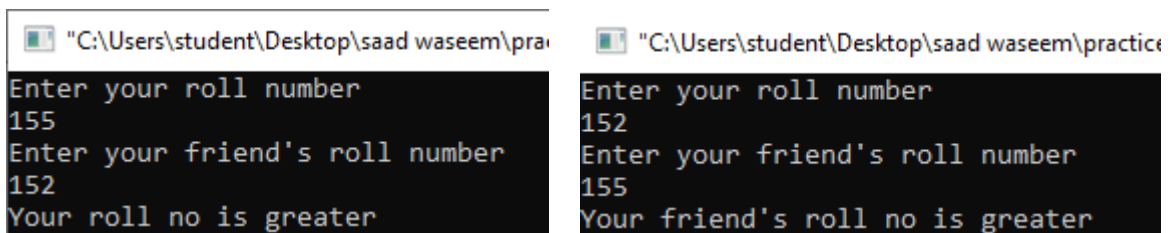
Task 3: Write a program that declares and initializes two numbers with your_roll_no and your_friend_roll_no and displays the greater of the two.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int your_roll_no, your_friend_roll_no;
    cout << "Enter your roll number\n";
    cin >> your_roll_no;
    cout << "Enter your friend's roll number\n";
    cin >> your_friend_roll_no;

    if (your_roll_no > your_friend_roll_no) {
        cout << "Your roll no is greater" << endl;
    }
    else {
        cout << "Your friend's roll no is greater" << endl;
    }
    return 0;
}
```

Output



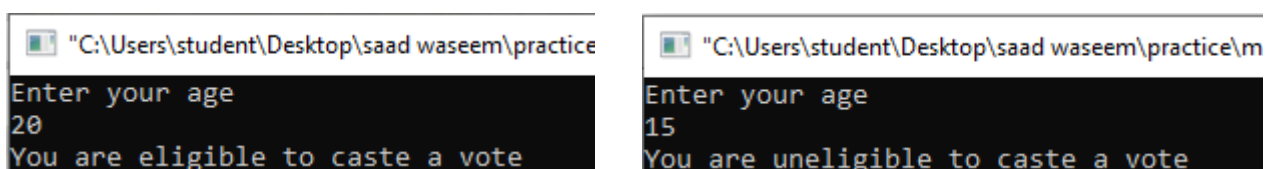
Task 4: Write a program to read the age of a candidate and determine whether it is eligible for casting his/her own vote.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int age;
    cout << "Enter your age\n";
    cin >> age;

    if (age >= 18) {
        cout << "You are eligible to caste a vote" << endl;
    }
    else {
        cout << "You are uneligible to caste a vote" << endl;
    }
    return 0;
}
```

Output



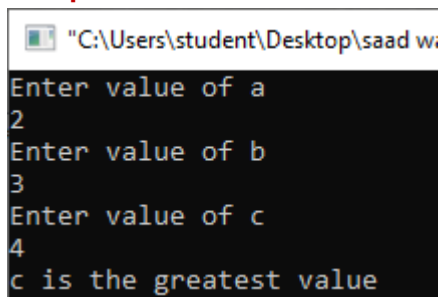
Task 5: Write a C++ program to enter three numbers from the user and display the greatest value among all three values.

Code

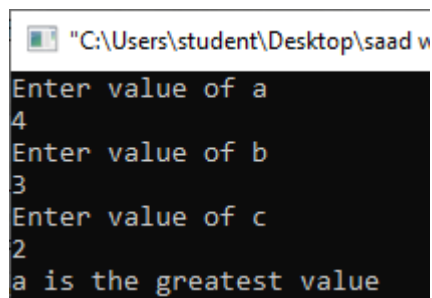
```
#include <iostream>
using namespace std;
int main()
{
    int a,b,c;
    cout << "Enter value of a\n";
    cin >> a;
    cout << "Enter value of b\n";
    cin >> b;
    cout << "Enter value of c\n";
    cin >> c;

    if (a > b && a > c) {
        cout << "a is the greatest value" << endl;
    }
    else if (b > a && b > c) {
        cout << "b is the greatest value" << endl;
    }
    else {
        cout << "c is the greatest value" << endl;
    }
    return 0;
}
```

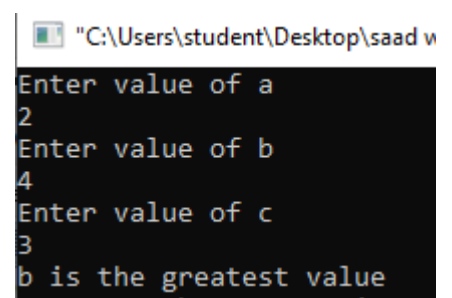
Output



```
"C:\Users\student\Desktop\saad w
Enter value of a
2
Enter value of b
3
Enter value of c
4
c is the greatest value
```



```
"C:\Users\student\Desktop\saad w
Enter value of a
4
Enter value of b
3
Enter value of c
2
a is the greatest value
```



```
"C:\Users\student\Desktop\saad w
Enter value of a
2
Enter value of b
4
Enter value of c
3
b is the greatest value
```

Lab Session 4 (Post-Lab Tasks)

Task 1: Input an alphabet (e.g. 'a', 'b', 'c', 'x', 'y', 'z') and display whether the input alphabet is a vowel (i.e. a, e, i, o, u) or not.

Code

```
#include <iostream>
using namespace std;
int main()
{
    char f;
    cout << "Enter an alphabet \n";
    cin >> f;

    if (f=='a' || f=='e' || f=='i' || f=='o' || f=='u') {
        cout << "The input alphabet is a vowel " << endl;
    }
    else if(f == 'A' || f == 'E' || f == 'I' || f == 'O' || f == 'U'){
        cout << "The input alphabet is a vowel " << endl;
    }
}
```



```

else{
    cout << "The input alphabet is not a vowel" << endl;
}
return 0;
}

```

Output

```

C:\> Microsoft Visual Studio Debug Console
Enter an alphabet
e
The input alphabet is a vowel

```

```

C:\> Microsoft Visual Studio Debug Console
Enter an alphabet
v
The input alphabet is not a vowel

```

```

C:\> Microsoft Visual Studio Debug Console
Enter an alphabet
A
The input alphabet is a vowel

```

```

C:\> Microsoft Visual Studio Debug Console
Enter an alphabet
X
The input alphabet is not a vowel

```

Task 2: Take three numbers as input from the user and display those numbers in descending order.

Code

```

#include<iostream>
using namespace std;
int main()
{
    int a, b, c;
    cout<<"Enter numbers: \n";
    cin>>a >> b >> c;

    if ((a >= b) && (a >= c))
    {
        if (b >= c)
        {
            cout << "Descending order: " << a << b << c << endl;
        }
        else
        {
            cout << "Descending order: " << a << c << b << endl;
        }
    }
    else if ((b >= a) && (b >= c))
    {
        if (a >= c)
        {
            cout << "Descending order: " << b << a << c << endl;
        }
        else
        {
            cout << "Descending order: " << b << c << a << endl;
        }
    }
    else if ((c >= a) && (c >= b))
    {
        if (a >= b)
        {
            cout << "Descending order: " << c << a << b << endl;
        }
    }
}

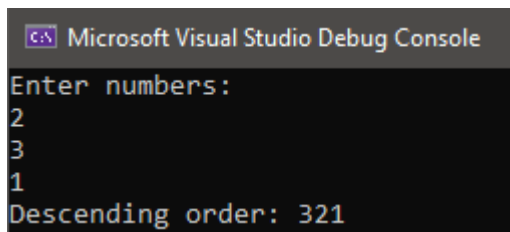
```

```

    else
    {
        cout << "Descending order: " << c << b << a << endl;
    }
}
}

```

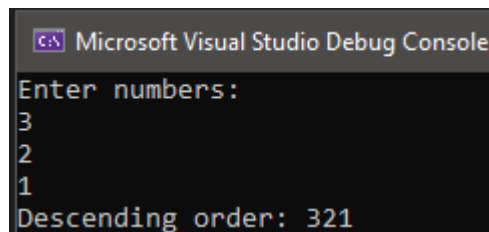
Output



```

Microsoft Visual Studio Debug Console
Enter numbers:
2
3
1
Descending order: 321

```



```

Microsoft Visual Studio Debug Console
Enter numbers:
3
2
1
Descending order: 321

```

Task 3: Read the Salary of an employee and make the following deductions before printing it out: If the salary is less than or equal to 10,000 then no deductions are done. If the salary is more than 10,000 and less than 25,000 then deduct Rs.1000. If the salary is equal to or more than 25,000 then deduct 5 % of the salary.

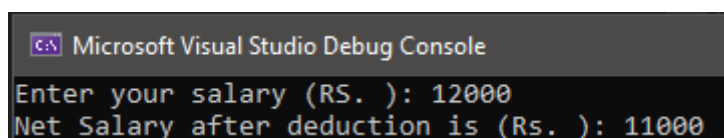
Code

```

#include <iostream>
using namespace std;
int main()
{
    int salary, r1, r3;
    float r2;
    cout << "Enter your salary (RS. ): ";
    cin >> salary;
    r1 = salary - 1000;
    r2 = ((5.0 / 100) * salary);
    r3 = salary - r2;
    if (salary <= 10000) {
        cout << "No deduction is done in the salary" << endl;
    }
    else if (salary > 10000 && salary < 25000) {
        cout << "Net Salary after deduction is (Rs. ): " << r1 << endl;
    }
    else if (salary >= 25000) {
        cout << "Net Salary after deduction is (Rs. ): " << r3 << endl;
    }
    return 0;
}

```

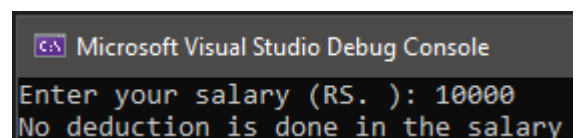
Output



```

Microsoft Visual Studio Debug Console
Enter your salary (RS. ): 12000
Net Salary after deduction is (Rs. ): 11000

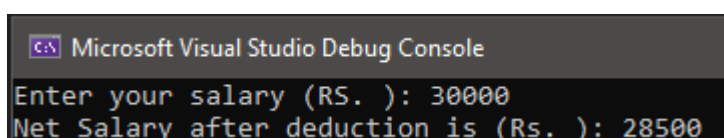
```



```

Microsoft Visual Studio Debug Console
Enter your salary (RS. ): 10000
No deduction is done in the salary

```



```

Microsoft Visual Studio Debug Console
Enter your salary (RS. ): 30000
Net Salary after deduction is (Rs. ): 28500

```

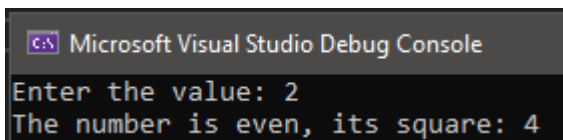
Task 4: Find out whether the input number is even or odd and if the number is even, display its square and if the number is odd display its cube.

Code

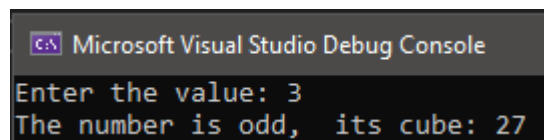
```
#include <iostream>
using namespace std;
int main()
{
    int i, x, y;
    cout << "Enter the value: ";
    cin >> i;
    x = i * i;
    y = i * i * i;

    if (i % 2 == 1) {
        cout << "The number is odd, its cube: " << y << endl;
    }
    else {
        cout << "The number is even, its square: " << x << endl;
    }
}
```

Output



```
Microsoft Visual Studio Debug Console
Enter the value: 2
The number is even, its square: 4
```



```
Microsoft Visual Studio Debug Console
Enter the value: 3
The number is odd, its cube: 27
```

Task 5: Ask the user for his/her age and display the message according to the following: If the age is less than 16, display "You can't drive." If the age is less than 18, display "You can't vote." If the age is less than 25, display "You can't rent a car." If the age is 25 or over, display "You can do anything that's legal."

Code

```
#include <iostream>
using namespace std;
int main()
{
    int age;
    cout << "Enter your age\n";
    cin >> age;

    if (age < 16) {
        cout << "You can't drive." << endl;
    }
    if (age < 18) {
        cout << "You can't vote." << endl;
    }
    if (age < 25) {
        cout << "You can't rent a car." << endl;
    }
    if (age >= 25) {
        cout << "You can do anything that's legal." << endl;
    }
    return 0;
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter your age
15
You can't drive.
You can't vote.
You can't rent a car.
```

```
Microsoft Visual Studio Debug Console
Enter your age
17
You can't vote.
You can't rent a car.
```

```
Microsoft Visual Studio Debug Console
Enter your age
30
You can do anything that's legal.
```

```
Microsoft Visual Studio Debug Console
Enter your age
24
You can't rent a car.
```

Task 6: If a five-digit number is an input through the keyboard, write a program to calculate the sum of its digits.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int num, sum, d1, d2, d3, d4, d5;
    cout << "Enter 5 digit value\n";
    cin >> num;
    if (num >= 10000 && num <= 99999) {
        d5 = num % 10;
        num = num / 10;
        d4 = num % 10;
        num = num / 10;
        d3 = num % 10;
        num = num / 10;
        d2 = num % 10;
        d1 = num / 10;
        sum = d5 + d4 + d3 + d2 + d1;
        cout << "Sum of digits: " << sum;
    }
    else {
        cout << "Number is invalid";
    }
    return 0;
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter 5 digit value
15215
Sum of digits: 14
```

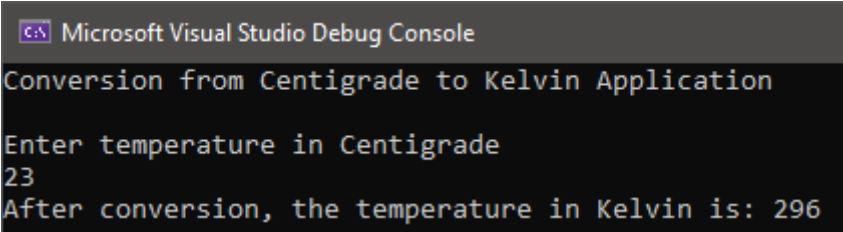
```
Microsoft Visual Studio Debug Console
Enter 5 digit value
12
Number is invalid
```

Task 6: Write a function that takes the temperature in Centigrade as an input argument and returns the Kelvin temperature

Code

```
#include <iostream>
using namespace std;
int main()
{
    cout << "Conversion from Centigrade to Kelvin Application\n";
    float c,k ;
    cout << "\nEnter temperature in Centigrade \n";
    cin >> c;
    k = c + 273;
    cout << "After conversion, the temperature in Kelvin is: " << k<<endl;
    return 0;
}
```

Output



The screenshot shows the Microsoft Visual Studio Debug Console with the following output:

```
Conversion from Centigrade to Kelvin Application

Enter temperature in Centigrade
23
After conversion, the temperature in Kelvin is: 296
```

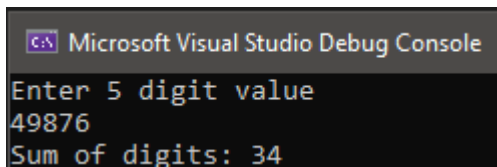
Lab Session 5 (In-Lab Tasks)

Task 1: Write and run a program that reads a 5-digit integer and prints the sum of its five digits.

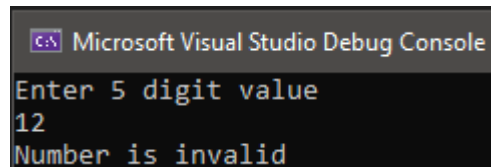
Code

```
#include <iostream>
using namespace std;
int main()
{
    int num, sum, d1, d2, d3, d4, d5;
    cout << "Enter 5 digit value\n";
    cin >> num;
    if (num >= 10000 && num <= 99999) {
        d5 = num % 10;
        num = num / 10;
        d4 = num % 10;
        num = num / 10;
        d3 = num % 10;
        num = num / 10;
        d2 = num % 10;
        d1 = num / 10;
        sum = d5 + d4 + d3 + d2 + d1;
        cout << "Sum of digits: " << sum;
    }
    else {
        cout << "Number is invalid";
    }
    return 0;
}
```

Output



```
Microsoft Visual Studio Debug Console
Enter 5 digit value
49876
Sum of digits: 34
```



```
Microsoft Visual Studio Debug Console
Enter 5 digit value
12
Number is invalid
```

Task 2: Write a program which takes 3 integers as inputs and find the 2nd maximum from three. Do it with help of && or || operators.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int a, b, c;
    cout << "Enter first number\n";
    cin >> a;
    cout << "Enter second number\n";
    cin >> b;
    cout << "Enter third number\n";
    cin >> c;
    if (a > b && a < c || a > c && a < b) {
        cout << "2nd maximum number from three numbers is: " << a;
    }
    else if (b > a && b < c || b > c && b < a) {
        cout << "2nd maximum number from three numbers is: " << b;
    }
    else if (c > a && c < b || c > b && c < a) {
        cout << "2nd maximum number from three numbers is: " << c;
    }
}
```

```

    }
    return 0;
}

```

Output

```

Microsoft Visual Studio Debug Console
Enter first number
3
Enter second number
2
Enter third number
1
2nd maximum number from three numbers is: 2

```

```

Microsoft Visual Studio Debug Console
Enter first number
9
Enter second number
12
Enter third number
8
2nd maximum number from three numbers is: 9

```

```

Microsoft Visual Studio Debug Console
Enter first number
30
Enter second number
55
Enter third number
45
2nd maximum number from three numbers is: 45

```

Task 3: Write and run a program that reads two integers and then uses the conditional expression operator to print either “multiple” or “not” according to whether one of the integers is a multiple of the other

Code

```

#include <iostream>
using namespace std;
int main()
{
    int a, b;
    cout << "Enter first number\n";
    cin >> a;
    cout << "Enter second number\n";
    cin >> b;

    if (a % b == 0) {
        cout << a << " is multiple of " << b << endl;
    }
    else{
        cout << a << " is not multiple of " << b << endl;
    }
    return 0;
}

```

Output

```

Microsoft Visual Studio Debug Console
Enter first number
50
Enter second number
25
50 is multiple of 25

```

```

Microsoft Visual Studio Debug Console
Enter first number
50
Enter second number
23
50 is not multiple of 23

```

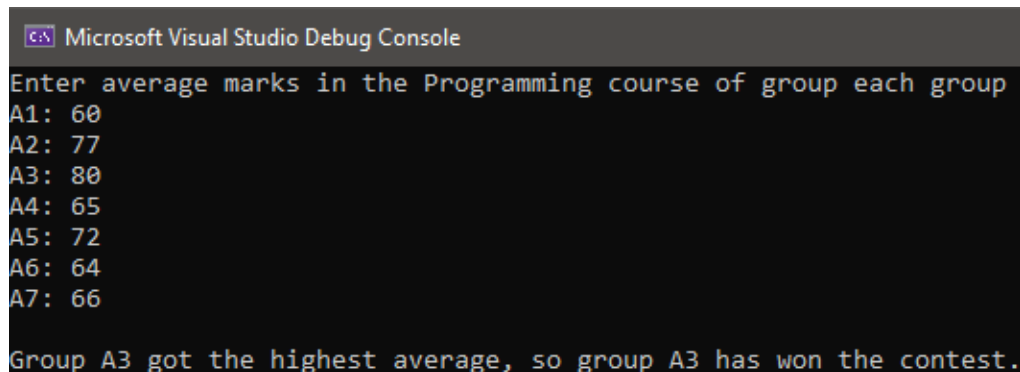
Task 4: Suppose your class has 7 groups. We are required to find out which group's average is higher. Write a program that takes average marks in the Programming course of each group and displays which group has won w.r.t average.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int A1, A2, A3, A4, A5, A6, A7;
    cout << "Enter average marks in the Programming course of group each group"<<endl;
    cout << "A1: ";
    cin >> A1;
    cout << "A2: ";
    cin >> A2;
    cout << "A3: ";
    cin >> A3;
    cout << "A4: ";
    cin >> A4;
    cout << "A5: ";
    cin >> A5;
    cout << "A6: ";
    cin >> A6;
    cout << "A7: ";
    cin >> A7;

    if ( A1>A2 && A1>A3 && A1>A4 && A1>A5 && A1>A6 && A1>A7 ) {
        cout << "\nGroup A1 got the highest average, so group A1 has won the contest.\n";
    }
    else if( A2 > A1 && A2 > A3 && A2 > A4 && A2 > A5 && A2 > A6 && A2 > A7){
        cout << "\nGroup A2 got the highest average, so group A2 has won the contest.\n";
    }
    else if (A3 > A1 && A3 > A2 && A3 > A4 && A3 > A5 && A3 > A6 && A3 > A7) {
        cout << "\nGroup A3 got the highest average, so group A3 has won the contest.\n";
    }
    else if (A4 > A1 && A4 > A2 && A4 > A3 && A4 > A5 && A4 > A6 && A4 > A7) {
        cout << "\nGroup A4 got the highest average, so group A4 has won the contest.\n";
    }
    else if (A5 > A1 && A5 > A2 && A5 > A4 && A5 > A3 && A5 > A6 && A5 > A7) {
        cout << "\nGroup A5 got the highest average, so group A5 has won the contest.\n";
    }
    else if (A6 > A1 && A6 > A2 && A6 > A4 && A6 > A5 && A6 > A3 && A6 > A7) {
        cout << "\nGroup A6 got the highest average, so group A6 has won the contest.\n";
    }
    else{
        cout << "\nGroup A7 got the highest average, so group A7 has won the contest.\n";
    }
    return 0;
}
```

Output



```
Microsoft Visual Studio Debug Console
Enter average marks in the Programming course of group each group
A1: 60
A2: 77
A3: 80
A4: 65
A5: 72
A6: 64
A7: 66

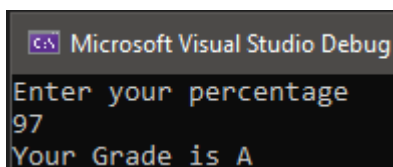
Group A3 got the highest average, so group A3 has won the contest.
```


Task 5: Write a C++ program that takes a percentage of a student from the user and prints the grade according to the percentage on the following criteria. The Grade is A if the Percentage is 80-100, the Grade is B if the Percentage is 70-79, Grade is C if the Percentage is 50-69, the Grade is F if the Percentage is 0-49, Error message if the percentage is above 100 or negative.

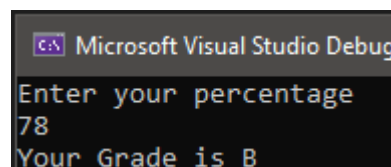
Code

```
#include <iostream>
using namespace std;
int main()
{
    int per;
    cout << "Enter your percentage"<<endl;
    cin >> per;
    if (per >= 0 && per <= 100){
        if (per >= 80 && per <= 100) {
            cout << "Your Grade is A" << endl;
        }
        else if (per >= 70 && per <= 79) {
            cout << "Your Grade is B" << endl;
        }
        else if (per >= 50 && per <= 69) {
            cout << "Your Grade is C" << endl;
        }
        else{
            cout << "Your Grade is F" << endl;
        }
    }
    else{
        cout << "Percentage entered is invalid\n";
    }
    return 0;
}
```

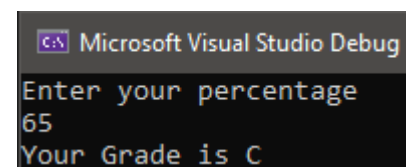
Output



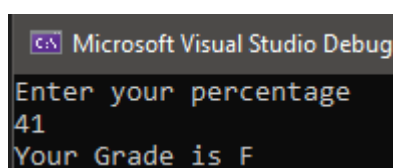
```
Microsoft Visual Studio Debug
Enter your percentage
97
Your Grade is A
```



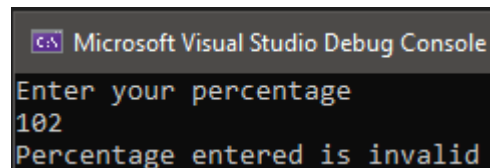
```
Microsoft Visual Studio Debug
Enter your percentage
78
Your Grade is B
```



```
Microsoft Visual Studio Debug
Enter your percentage
65
Your Grade is C
```



```
Microsoft Visual Studio Debug
Enter your percentage
41
Your Grade is F
```



```
Microsoft Visual Studio Debug Console
Enter your percentage
102
Percentage entered is invalid
```

Lab Session 5 (Post-Lab Tasks)

Task 1: Write a program that takes the marks of 5 different students (1,2,3,4,5) in 5 different courses as input and outputs the highest marks in each course along with the id of the student(e.g. 1,2,3,4,5) who scored it.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int S1, S2, S3, S4, S5;
    float marks, S1M, S1P, S1C, S1E, S1U;
    float S2M, S2P, S2C, S2E, S2U;
    float S3M, S3P, S3C, S3E, S3U;
    float S4M, S4P, S4C, S4E, S4U;
```

```
float S5M, S5P, S5C, S5E, S5U;
```

```
    cout << "Enter marks of Math, Physics, Computer, English, and Urdu courses  
respectively of the student having ID no. S1:" << endl;  
    cin >> S1M >> S1P >> S1C >> S1E >> S1U;  
    cout << "Enter marks of Math, Physics, Computer, English, and Urdu courses  
respectively of the student having ID no. S2:" << endl;  
    cin >> S2M >> S2P >> S2C >> S2E >> S2U;  
    cout << "Enter marks of Math, Physics, Computer, English, and Urdu courses  
respectively of the student having ID no. S3:" << endl;  
    cin >> S3M >> S3P >> S3C >> S3E >> S3U;  
    cout << "Enter marks of Math, Physics, Computer, English, and Urdu courses  
respectively of the student having ID no. S4:" << endl;  
    cin >> S4M >> S4P >> S4C >> S4E >> S4U;  
    cout << "Enter marks of Math, Physics, Computer, English, and Urdu courses  
respectively of the student having ID no. S5:" << endl;  
    cin >> S5M >> S5P >> S5C >> S5E >> S5U;
```

```
    if (S1M > S2M && S1M > S3M && S1M > S4M && S1M > S5M) {  
        cout << "Student having an ID no. S1 has the highest marks in the Math  
course that is: " << S1M << endl;  
    }  
    else if (S2M > S1M && S2M > S3M && S2M > S4M && S2M > S5M) {  
        cout << "Student having an ID no. S2 has the highest marks in the Math  
course that is: " << S2M << endl;  
    }  
    else if (S3M > S1M && S3M > S2M && S3M > S4M && S3M > S5M) {  
        cout << "Student having an ID no. S3 has the highest marks in the Math  
course that is: " << S3M << endl;  
    }  
    else if (S4M > S1M && S4M > S2M && S4M > S3M && S4M > S5M) {  
        cout << "Student having an ID no. S4 has the highest marks in the Math  
course that is: " << S4M << endl;  
    }  
    else {  
        cout << "Student having an ID no. S5 has the highest marks in the Math  
course that is: " << S5M << endl;  
    }
```

```
    if (S1P > S2P && S1P > S3P && S1P > S4P && S1P > S5P) {  
        cout << "Student having an ID no. S1 has the highest marks in the Physics  
course that is: " << S1P << endl;  
    }  
    else if (S2P > S1P && S2P > S3P && S2P > S4P && S2P > S5P) {  
        cout << "Student having an ID no. S2 has the highest marks in the Physics  
course that is: " << S2P << endl;  
    }  
    else if (S3P > S1P && S3P > S2P && S3P > S4P && S3P > S5P) {  
        cout << "Student having an ID no. S3 has the highest marks in the Physics  
course that is: " << S3P << endl;  
    }  
    else if (S4P > S1P && S4P > S2P && S4P > S3P && S4P > S5P) {  
        cout << "Student having an ID no. S4 has the highest marks in the Physics  
course that is: " << S4P << endl;  
    }  
    else {  
        cout << "Student having an ID no. S5 has the highest marks in the Physics  
course that is: " << S5P << endl;  
    }
```

```
    if (S1C > S2C && S1C > S3C && S1C > S4C && S1C > S5C) {  
        cout << "Student having an ID no. S1 has the highest marks in the Computer  
course that is: " << S1C << endl;
```

```

    }
    else if (S2C > S1C && S2C > S3C && S2C > S4C && S2C > S5C) {
        cout << "Student having an ID no. S2 has the highest marks in the Computer
course that is: " << S2C << endl;
    }
    else if (S3C > S1C && S3C > S2C && S3C > S4C && S2C > S5C) {
        cout << "Student having an ID no. S3 has the highest marks in the Computer
course that is: " << S3C << endl;
    }
    else if (S4C > S1C && S2C > S3C && S2C > S4C && S2C > S5C) {
        cout << "Student having an ID no. S4 has the highest marks in the Computer
course that is: " << S4C << endl;
    }
    else {
        cout << "Student having an ID no. S5 has the highest marks in the Computer
course that is: " << S5C << endl;
    }

    if (S1E > S2E && S1E > S3E && S1E > S4E && S1E > S5E) {
        cout << "Student having an ID no. S1 has the highest marks in the English
course that is: " << S1E << endl;
    }
    else if (S2E > S1E && S2E > S3E && S2E > S4E && S2E > S5E) {
        cout << "Student having an ID no. S2 has the highest marks in the English
course that is: " << S2E << endl;
    }
    else if (S3E > S1E && S3E > S2E && S3E > S4E && S2E > S5E) {
        cout << "Student having an ID no. S3 has the highest marks in the English
course that is: " << S3E << endl;
    }
    else if (S4E > S1E && S2E > S3E && S2E > S4E && S2E > S5E) {
        cout << "Student having an ID no. S4 has the highest marks in the English
course that is: " << S4E << endl;
    }
    else {
        cout << "Student having an ID no. S5 has the highest marks in the English
course that is: " << S5E << endl;
    }

    if (S1U > S2U && S1U > S3U && S1U > S4U && S1U > S5U) {
        cout << "Student having an ID no. S1 has the highest marks in the Urdu
course that is: " << S1U << endl;
    }
    else if (S2U > S1U && S2U > S3U && S2U > S4U && S2U > S5U) {
        cout << "Student having an ID no. S2 has the highest marks in the Urdu
course that is: " << S2U << endl;
    }
    else if (S3U > S1U && S3U > S2U && S3U > S4U && S2U > S5U) {
        cout << "Student having an ID no. S3 has the highest marks in the Urdu
course that is: " << S3U << endl;
    }
    else if (S4U > S1U && S2U > S3U && S2U > S4U && S2U > S5U) {
        cout << "Student having an ID no. S4 has the highest marks in the Urdu
course that is: " << S4U << endl;
    }
    else {
        cout << "Student having an ID no. S5 has the highest marks in the Urdu
course that is: " << S5U << endl;
    }
    return 0;
}

```

Output

```
Microsoft Visual Studio Debug Console
Enter marks of Math, Physics, Computer, English, and Urdu courses respectively of the student having ID no. S1:
77
76
75
76
85
Enter marks of Math, Physics, Computer, English, and Urdu courses respectively of the student having ID no. S2:
94
98
78
77
56
Enter marks of Math, Physics, Computer, English, and Urdu courses respectively of the student having ID no. S3:
89
88
61
53
76
Enter marks of Math, Physics, Computer, English, and Urdu courses respectively of the student having ID no. S4:
62
60
50
90
93
Enter marks of Math, Physics, Computer, English, and Urdu courses respectively of the student having ID no. S5
85
70
87
85
77
Student having an ID no. S2 has the highest marks in the Math course that is: 94
Student having an ID no. S2 has the highest marks in the Physics course that is: 98
Student having an ID no. S5 has the highest marks in the Computer course that is: 87
Student having an ID no. S5 has the highest marks in the English course that is: 85
Student having an ID no. S5 has the highest marks in the Urdu course that is: 77
```

Task 2: Create a BMI calculator application that reads the user's weight in pounds and height in inches (or, if you prefer, the user's weight in kilograms and height in meters), then calculates and displays the user's body mass index based on following values.

Code

```
#include<iostream>
using namespace std;
int main()
{
    float bmi, w, h;
    cout << "Please Enter your weight in Pounds: ";
    cin >> w;
    cout << "Please Enter your Height in Inches: ";
    cin >> h;
    bmi = (w * 703) / (h * h);
    cout << "Your BMI is: " << bmi << endl;

    if (bmi < 18.5){
        cout << "Underweight";
    }
    else if (bmi <= 24.9 && bmi >= 18.5){
        cout << "Normal";
    }
    else if(bmi <= 29.9 && bmi >= 25){
        cout << "Overweight";
    }
    else{
        cout << "Obese";
    }
    return 0;
}
```

```
}
```

Output

```
Microsoft Visual Studio Debug Console
Please Enter your weight in Pounds: 143.3
Please Enter your Height in Inches: 69.1
Your BMI is: 21.0982
Normal
```

```
Microsoft Visual Studio Debug Console
Please Enter your weight in Pounds: 77.16
Please Enter your Height in Inches: 69.1
Your BMI is: 11.3603
Underweight
```

```
Microsoft Visual Studio Debug Console
Please Enter your weight in Pounds: 264.55
Please Enter your Height in Inches: 69.1
Your BMI is: 38.95
Obese
```

```
Microsoft Visual Studio Debug Console
Please Enter your weight in Pounds: 198.41
Please Enter your Height in Inches: 69.1
Your BMI is: 29.2121
Overweight
```

Task 3: Write and run a program that simulates a simple calculator. It reads two integers and a character. If the character is a '+', the sum of integers is printed; if the character is a '-', the subtraction of integers is printed; if it is a '*', the product is printed; if it is a '/', the quotient is printed; and if it is a '%', the remainder is printed.

Code

```
#include <iostream>
using namespace std;
int main()
{
    cout << "Calculator Application\n";
    int a, b, sum, sub, multiply, divide, remainder;
    bool o;
    cout << "\nEnter first Value: ";
    cin >> a;
    cout << "Enter second value: ";
    cin >> b;
    cout << "Enter operator: ";
    cin >> o;

    if (o == "+") {
        sum = a + b;
        cout << "Result of addition is: " << sum;
    }
    else if (o == '-') {
        sub = a - b;
        cout << "Result of subtraction is: " << sub;
    }
    else if (o == '*') {
        multiply = a * b;
        cout << "Result of multiplication is: " << multiply;
    }
    else if (o == '/') {
        divide = a / b;
        cout << "Result of division is: " << divide ;
    }
    else if (o == '%') {
        remainder = a % b;
        cout << "The remainder is: " << remainder;
    }
    return 0;
}
```

Output

```
Microsoft Visual Studio Debug Console
Calculator Application

Enter first Value: 10
Enter socond value: 5
Enter operator: +
Result of addition is: 15
```

```
Microsoft Visual Studio Debug Console
Calculator Application

Enter first Value: 12
Enter socond value: 2
Enter operator: -
Result of subtraction is: 10
```

```
Microsoft Visual Studio Debug Console
Calculator Application

Enter first Value: 15
Enter socond value: 5
Enter operator: /
Result of division is: 3
```

```
Microsoft Visual Studio Debug Console
Calculator Application

Enter first Value: 50
Enter socond value: 11
Enter operator: %
The remainder is: 6
```

Task 4: Write a program which takes marks of 5 courses as input, of 5 students and output the students who got the highest aggregate.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int S1, S2, S3, S4, S5;
    float marks,a1,a2,a3,a4,a5,S1M, S1P, S1C, S1E, S1U;
    float S2M, S2P, S2C, S2E, S2U;
    float S3M, S3P, S3C, S3E, S3U;
    float S4M, S4P, S4C, S4E, S4U;
    float S5M, S5P, S5C, S5E, S5U;
    cout << "Enter marks of all courses of the student having Roll no. S1" << endl;
    cin >> S1M >> S1P >> S1C >> S1E >> S1U;

    cout << "Enter marks of all courses of the student having Roll no. S2" << endl;
    cin >> S2M >> S2P >> S2C >> S2E >> S2U;

    cout << "Enter marks of all courses of the student having Roll no. S3" << endl;
    cin >> S3M >> S3P >> S3C >> S3E >> S3U;

    cout << "Enter marks of all courses of the student having Roll no. S4" << endl;
    cin >> S4M >> S4P >> S4C >> S4E >> S4U;

    cout << "Enter marks of all courses of the student having Roll no. S5" << endl;
    cin >> S5M >> S5P >> S5C >> S5E >> S5U;

    a1 = S1M + S1P + S1C + S1E + S1U;
    a2 = S2M + S2P + S2C + S2E + S2U;
    a3 = S3M + S3P + S3C + S3E + S3U;
    a4 = S4M + S4P + S4C + S4E + S4U;
    a5 = S5M + S5P + S5C + S5E + S5U;

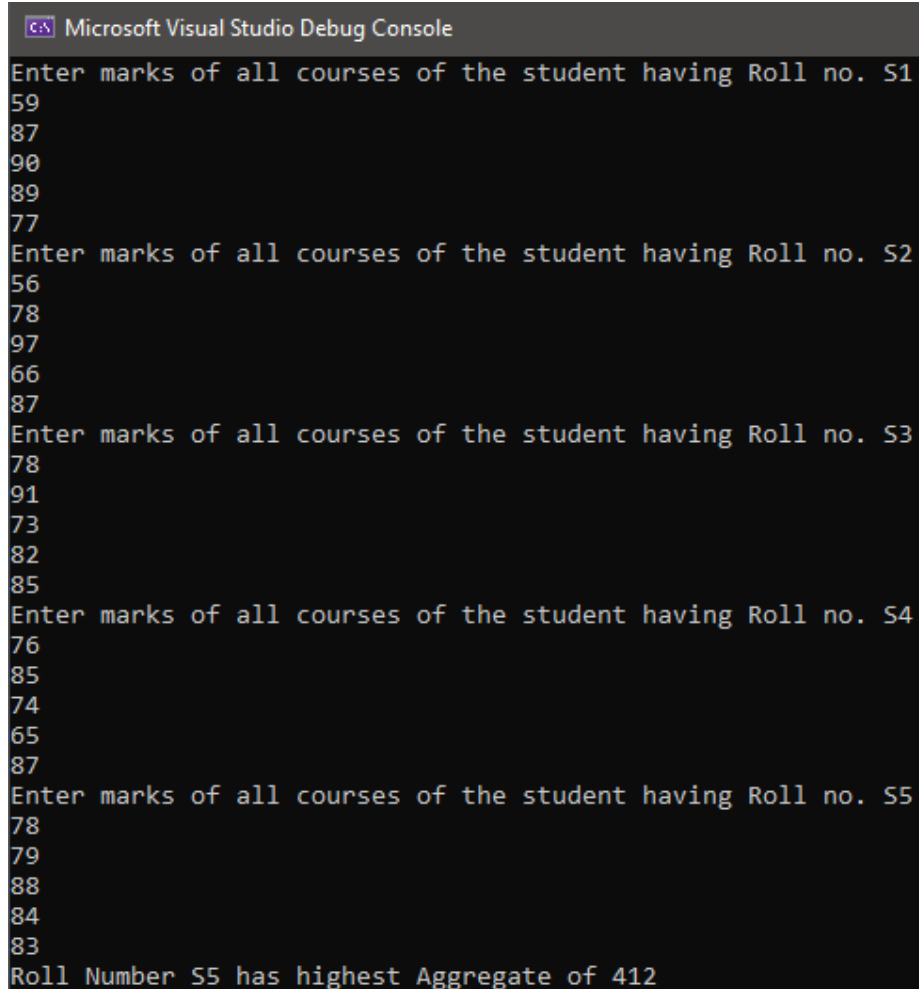
    if (a1>a2 && a1>a3 && a1>a4 && a1>a5) {
        cout << "Roll Number S1 has highest Aggregate of "<<a1<< endl;
    }
    else if (a2 > a1 && a2 > a3 && a2 > a4 && a2 > a5) {
        cout << "Roll Number S2 has highest Aggregate of " << a2 << endl;
    }
    else if (a3 > a1 && a3 > a2 && a3 > a4 && a3 > a5) {
        cout << "Roll Number S3 has highest Aggregate of " << a3 << endl;
    }
    else if (a4 > a1 && a4 > a3 && a4 > a2 && a4 > a5) {
```

```

        cout << "Roll Number S4 has highest Aggregate of " << a4 << endl;
    }
    else {
        cout << "Roll Number S5 has highest Aggregate of " << a5 << endl;
    }
    return 0;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter marks of all courses of the student having Roll no. S1
59
87
90
89
77
Enter marks of all courses of the student having Roll no. S2
56
78
97
66
87
Enter marks of all courses of the student having Roll no. S3
78
91
73
82
85
Enter marks of all courses of the student having Roll no. S4
76
85
74
65
87
Enter marks of all courses of the student having Roll no. S5
78
79
88
84
83
Roll Number S5 has highest Aggregate of 412

```

Lab Task 5

Task 1: Prompt the user for the diameter of a circle in centimeters and find out its area in square meters.

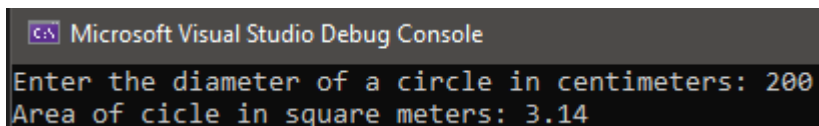
Code

```

#include <iostream>
using namespace std;
int main()
{
    float dc, dm, r, a;
    cout << "Enter the diameter of a circle in centimeters: ";
    cin >> dc;
    dm = dc * 0.01;
    r = dm / 2;
    a = 3.14 * (r * r);
    cout << "Area of circle in square meters: " << a << endl;
    return 0;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter the diameter of a circle in centimeters: 200
Area of circle in square meters: 3.14

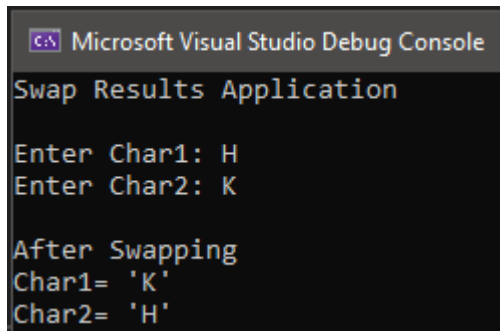
```

Task 2: Input two characters char1 and char2 and interchange their values i.e. swap.

Code

```
#include <iostream>
using namespace std;
int main()
{
    cout << "Swap Results Application\n";
    char char1, char2, temporary;
    cout << "\nEnter Char1: ";
    cin >> char1;
    cout << "Enter Char2: ";
    cin >> char2;
    temporary = char1;
    char1 = char2;
    char2 = temporary;
    cout << "\nAfter Swapping" << endl;
    cout << "Char1= " << "'" << char1 << "'" << endl;
    cout << "Char2= " << "'" << char2 << "'" << endl;
    return 0;
}
```

Output

A screenshot of the Microsoft Visual Studio Debug Console. The window title is "Microsoft Visual Studio Debug Console". The output text is as follows:
Swap Results Application

Enter Char1: H
Enter Char2: K

After Swapping
Char1= 'K'
Char2= 'H'

Task 3: If cost price and selling price of an item is input through the keyboard, write a program to determine whether the seller has made profit or incurred loss. Also determine how much profit he made or loss he incurred.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int sp,cp,profit,loss;
    cout << "Enter Cost price: ";
    cin >> cp;
    cout << "Enter selling price: ";
    cin >> sp;

    if (sp > cp) {
        profit = sp - cp;
        cout << "The seller has made profit of (Rs.): " << profit << endl;
    }
    else {
        loss = cp - sp;
        cout << "The seller has made loss of (Rs.): " << loss << endl;
    }
    return 0;
}
```


Output

```
Microsoft Visual Studio Debug Console
Enter Cost price: 500
Enter selling price: 800
The seller has made profit of (Rs.): 300
```

```
Microsoft Visual Studio Debug Console
Enter Cost price: 900
Enter selling price: 700
The seller has made loss of (Rs.): 200
```

Task 4: Write a program to check whether a triangle is valid or not, when the three angles of the triangle are entered through the keyboard. A triangle is valid if the sum of all three angles is equal to 180 degrees.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int a1,a2,a3,sum;
    cout << "Enter the three angles of the triangle:" << endl;
    cin >> a1 >> a2 >> a3;
    sum = a1 + a2 + a3;

    if (sum == 180) {
        cout << "Triangle is valid" << endl;
    }
    else {
        cout << "Triangle is invalid" << endl;
    }
    return 0;
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter the three angles of the triangle:
40
55
40
Triangle is invalid
```

```
Microsoft Visual Studio Debug Console
Enter the three angles of the triangle:
50
50
80
Triangle is valid
```

Lab Session 6 (In-Lab Tasks)

Task 1: What would be the output of the following program?

```
void main()
{
    int i = 2;
    switch (i)
    {
    case 1:
        cout << "I am in case 1 \n" << endl;
    case 2:
        cout << "I am in case 2 \n" << endl;
    case 3:
        cout << "I am in case 3 \n" << endl;
    default:
        cout << "I am in default \n" << endl;
    }
}
```

Errors

- `#include<iostream>` is missing. `#include<iostream>` tells the processor to include the contents of the file `<iostream>`
- `std::` is missing before `cout`. We need `std::` to specify what “namespace” `cout` belongs to.
- return type of ‘main’ should be ‘int’ instead of ‘void’ because the `void main()` indicates that the `main()` function will not return any value, but the `int main()` indicates that the `main()` can return integer type data.

Output

I am in case 2

I am in case 3

I am in default

Task 2: What would be the output of following program?

```
#include <iostream.h>
void main()
{
    int a = 10;
    switch (a) {
        case 5 + 5:
            cout << "Hello\n";
        default:
            cout << "OK\n";
    }
}
```

Errors

- `std::` is missing before `cout`. We need `std::` to specify what “namespace” `cout` belongs to.
- return type of ‘main’ should be ‘int’ instead of ‘void’ because the `void main()` indicates that the `main()` function will not return any value, but the `int main()` indicates that the `main()` can return integer type data.
- Missing closing quote before Hello.

Output

Hello

OK

Task 3: Write a C++ program to find maximum between two numbers using switch case.

Code

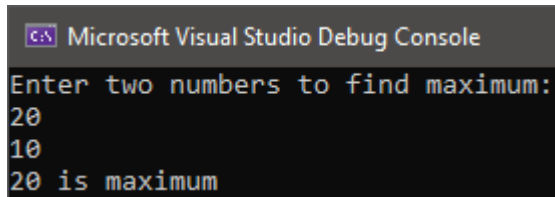
```
#include <iostream>
using namespace std;
int main()
{
    int a, b;
    cout << "Enter two numbers to find maximum: " << endl;
    cin >> a >> b;
    switch (a > b)
    {
        case 0:
            cout << b << " is maximum" << endl;
```

```

        break;
    case 1:
        cout << a << " is maximum" << endl;
        break;
    }
}

```

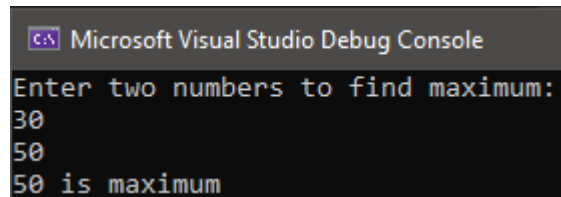
Output



```

Microsoft Visual Studio Debug Console
Enter two numbers to find maximum:
20
10
20 is maximum

```



```

Microsoft Visual Studio Debug Console
Enter two numbers to find maximum:
30
50
50 is maximum

```

Task 4. Write a C++ program to check whether a number is even or odd using switch case.

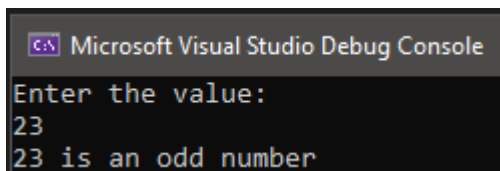
Code

```

#include <iostream>
using namespace std;
int main()
{
    int a;
    cout << "Enter the value: " << endl;
    cin >> a;
    switch ( a % 2 )
    {
        case 0:
            cout << a << " is an even number" << endl;
            break;
        case 1:
            cout << a << " is an odd number" << endl;
            break;
    }
}

```

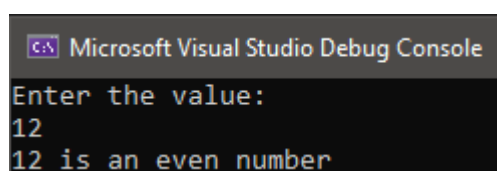
Output



```

Microsoft Visual Studio Debug Console
Enter the value:
23
23 is an odd number

```



```

Microsoft Visual Studio Debug Console
Enter the value:
12
12 is an even number

```

Task 5: Write a C++ program to check whether a number is positive or negative using a switch case.

Code

```

#include <iostream>
using namespace std;
int main()
{
    int a;
    cout << "Enter the value: " << endl;
    cin >> a;
    switch (a >= 1)
    {
        case 1:
            cout << a << " is a positive number" << endl;
            break;
    }
}

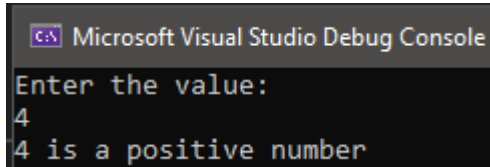
```

```

    case 0:
        switch (a <= -1) {
            case 1:
                cout << a << " is a negative number" << endl;
                break;
            }
        break;
    }
}

```

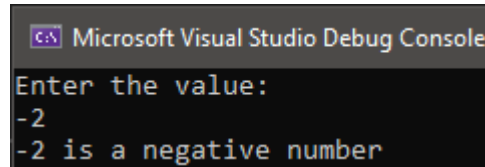
Output



```

Microsoft Visual Studio Debug Console
Enter the value:
4
4 is a positive number

```



```

Microsoft Visual Studio Debug Console
Enter the value:
-2
-2 is a negative number

```

Task 6. Write a program to create a Simple Calculator using a switch case.

Code

```

#include <iostream>
using namespace std;
int main()
{
    cout << "Calculator Application\n";
    int a, b, sum, sub, multiply, divide, choice;

    cout << "\nChoose which operation you want to execute:" << endl;
    cout << "1. Addition" << endl;
    cout << "2. Subtraction" << endl;
    cout << "3. Multiplication" << endl;
    cout << "4. Division" << endl;
    cin >> choice;

    switch (choice) {
        case 1:
            cout << "\nEnter first Value: ";
            cin >> a;
            cout << "Enter socond value: ";
            cin >> b;
            sum = a + b;
            cout << "Result of addition is: " << sum << endl;
            break;
        case 2:
            cout << "\nEnter first Value: ";
            cin >> a;
            cout << "Enter socond value: ";
            cin >> b;
            sub = a - b;
            cout << "Result of Subtraction is: " << sub << endl;
            break;
        case 3:
            cout << "\nEnter first Value: ";
            cin >> a;
            cout << "Enter socond value: ";
            cin >> b;
            multiply = a * b;
            cout << "Result of multiplication is: " << multiply << endl;
            break;
    }
}

```

```

        case 4:
            cout << "\nEnter first Value: ";
            cin >> a;
            cout << "Enter socond value: ";
            cin >> b;
            divide = a / b;
            cout << "Result of division is: " << divide << endl;
            break;
    }
    return 0;
}

```

Output

```

Microsoft Visual Studio Debug Console
Calculator Application

Choose which operation you want to execute:
1. Addition
2. Subtraction
3. Multiplication
4. Division
1

Enter first Value: 5
Enter socond value: 5
Result of addition is: 10

```

```

Microsoft Visual Studio Debug Console
Calculator Application

Choose which operation you want to execute:
1. Addition
2. Subtraction
3. Multiplication
4. Division
4

Enter first Value: 12
Enter socond value: 2
Result of division is: 6

```

```

Microsoft Visual Studio Debug Console
Calculator Application

Choose which operation you want to execute:
1. Addition
2. Subtraction
3. Multiplication
4. Division
3

Enter first Value: 2
Enter socond value: 2
Result of multiplication is: 4

```

```

Microsoft Visual Studio Debug Console
Calculator Application

Choose which operation you want to execute:
1. Addition
2. Subtraction
3. Multiplication
4. Division
2

Enter first Value: 19
Enter socond value: 3

```

Lab Session 6 (post-Lab Tasks)

Task 1: Write a program to print the day of the week name using a switch case.

Code

```

#include <iostream>
using namespace std;
int main()
{
    int week;
    cout << "Enter week number (1-7): ";
    cin >> week;

    switch (week) {
        case 1:
            cout << "Monday" << endl;
            break;
        case 2:
            cout << "Tuesday" << endl;
            break;
        case 3:
            cout << "Wednesday" << endl;
            break;
        case 4:
            cout << "Thursday" << endl;

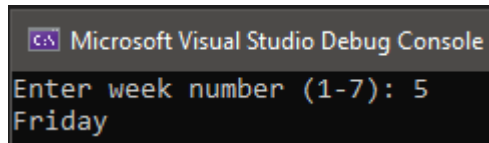
```

```

        break;
    case 5:
        cout << "Friday" << endl;
        break;
    case 6:
        cout << "Saturday" << endl;
        break;
    case 7:
        cout << "Sunday" << endl;
        break;
    default:
        cout << "You have entered an invalid number" << endl;
    }
    return 0;
}

```

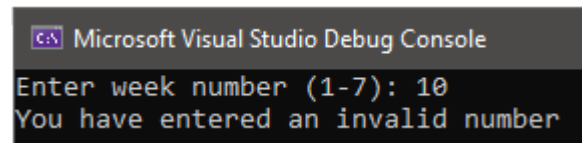
Output



```

Microsoft Visual Studio Debug Console
Enter week number (1-7): 5
Friday

```



```

Microsoft Visual Studio Debug Console
Enter week number (1-7): 10
You have entered an invalid number

```

Task 2: Write a program to print the total number of days in a month using a switch case.

Code

```

#include <iostream>
using namespace std;
int main()
{
    int month;
    cout << "Enter month number(1-12): ";
    cin >> month;

    switch (month) {
        case 1:
            cout << "January has 31 days" << endl;
            break;
        case 2:
            cout << "Febraury has 28/29 days" << endl;
            break;
        case 3:
            cout << "March has 31 days" << endl;
            break;
        case 4:
            cout << "April has 30 days" << endl;
            break;
        case 5:
            cout << "May has 31 days" << endl;
            break;
        case 6:
            cout << "June has 30 days" << endl;
            break;
        case 7:
            cout << "July has 31 days" << endl;
            break;
        case 8:
            cout << "August has 31 days" << endl;
            break;
        case 9:
            cout << "September has 30 days" << endl;

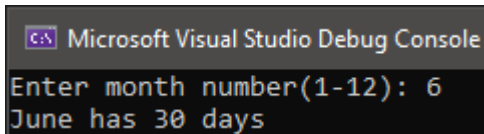
```

```

        break;
    case 10:
        cout << "October has 31 days" << endl;
        break;
    case 11:
        cout << "November has 30 days" << endl;
        break;
    case 12:
        cout << "December has 31 days" << endl;
        break;
    default:
        cout << "You have entered an invalid number" << endl;
}
return 0;
}

```

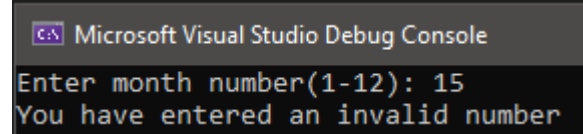
Output



```

C:\> Microsoft Visual Studio Debug Console
Enter month number(1-12): 6
June has 30 days

```



```

C:\> Microsoft Visual Studio Debug Console
Enter month number(1-12): 15
You have entered an invalid number

```

Task 3: Write a C program to check whether an alphabet is vowel or consonant using switch case.

Code

```

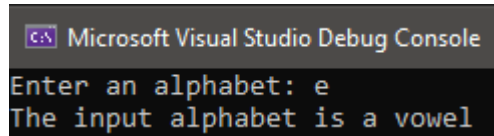
#include <iostream>
using namespace std;
int main()
{
    char f;
    cout << "Enter an alphabet: ";
    cin >> f;

    switch (f) {
        case 'a':
            cout << "The input alphabet is a vowel " << endl;
            break;
        case 'e':
            cout << "The input alphabet is a vowel " << endl;
            break;
        case 'i':
            cout << "The input alphabet is a vowel " << endl;
            break;
        case 'o':
            cout << "The input alphabet is a vowel " << endl;
            break;
        case 'u':
            cout << "The input alphabet is a vowel " << endl;
            break;
        case 'A':
            cout << "The input alphabet is a vowel " << endl;
            break;
        case 'E':
            cout << "The input alphabet is a vowel " << endl;
            break;
    }
}

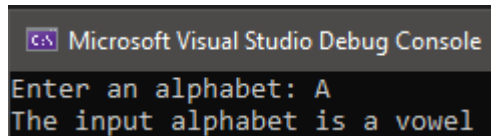
```

```
case 'I':  
    cout << "The input alphabet is a vowel " << endl;  
    break;  
case 'O':  
    cout << "The input alphabet is a vowel " << endl;  
    break;  
case 'U':  
    cout << "The input alphabet is a vowel " << endl;  
    break;  
default:  
    cout << "The input alphabet is a consonant " << endl;  
}  
return 0;
```

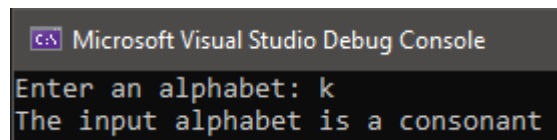
Output



Microsoft Visual Studio Debug Console
Enter an alphabet: e
The input alphabet is a vowel



Microsoft Visual Studio Debug Console
Enter an alphabet: A
The input alphabet is a vowel



Microsoft Visual Studio Debug Console
Enter an alphabet: k
The input alphabet is a consonant

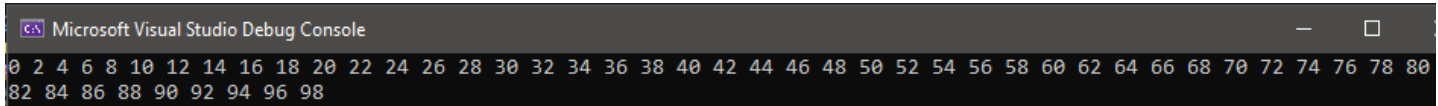
Lab Session 7 (In-Lab Tasks)

Task 1: Write a program that prints the first 50 even numbers by using for loop.

Code

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 0; i < 100; i += 2) {
        cout << i << " ";
    }
}
```

Output



Microsoft Visual Studio Debug Console

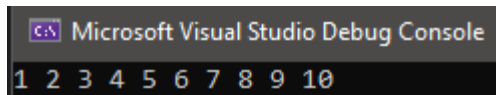
0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96 98

Task 2: Write a C++ program that will print the first 10 natural numbers by using a while loop

Code

```
#include <iostream>
using namespace std;
int main() {
    int i = 1;
    while (i <= 10) {
        cout << i << " ";
        i++;
    }
}
```

Output



Microsoft Visual Studio Debug Console

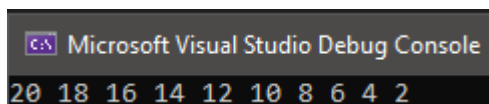
1 2 3 4 5 6 7 8 9 10

Task 3: Write a C++ program which will print even numbers from 20 to 1 by using for loop.

Code

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 20; i >= 1; i -= 2) {
        cout << i << " ";
    }
}
```

Output



Microsoft Visual Studio Debug Console

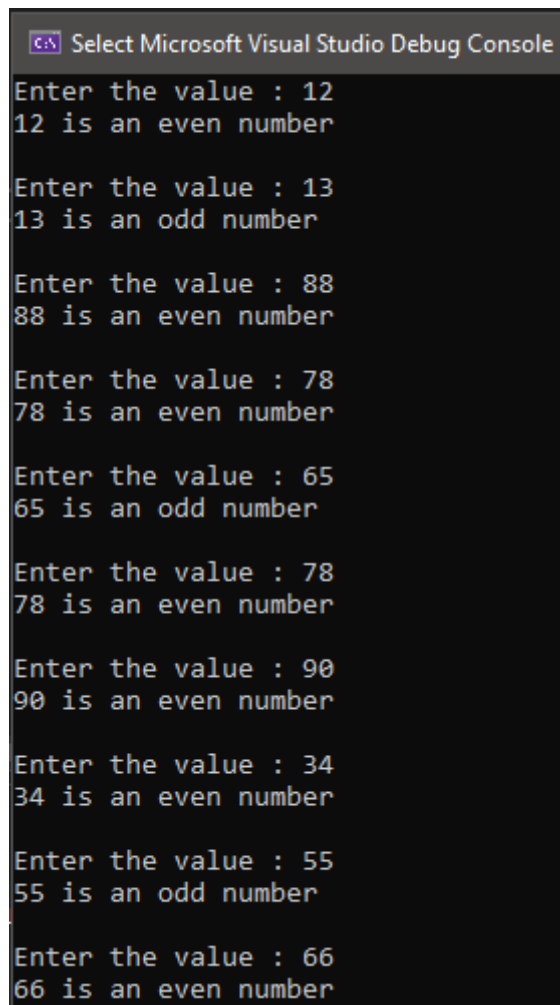
20 18 16 14 12 10 8 6 4 2

Task 4: Write a C++ program which will take ten numbers from user and for each number display whether the given number is even or odd.

Code

```
#include <iostream>
using namespace std;
int main(){
    int i;
    int X = 1;
    while (X<= 10) {
        cout <<"\nEnter the value : ";
        cin >> i;
        if (i % 2 == 1) {
            cout << i << " is an odd number" << endl;
        }
        else
            cout << i << " is an even number" << endl;
        X++;
    }
}
```

Output



```
Select Microsoft Visual Studio Debug Console

Enter the value : 12
12 is an even number

Enter the value : 13
13 is an odd number

Enter the value : 88
88 is an even number

Enter the value : 78
78 is an even number

Enter the value : 65
65 is an odd number

Enter the value : 78
78 is an even number

Enter the value : 90
90 is an even number

Enter the value : 34
34 is an even number

Enter the value : 55
55 is an odd number

Enter the value : 66
66 is an even number
```

Task 5: Write a C++ program which will print first 10 numbers which are multiple of 5.

Code

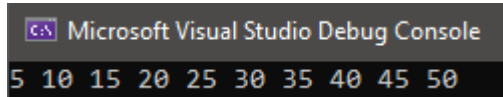
```
#include <iostream>
using namespace std;
int main() {
```

```

for (int i=1; i <= 50; i++) {
    if (i % 5 == 0) {
        cout << i << " ";
    }
}
}

```

Output



```

Microsoft Visual Studio Debug Console
5 10 15 20 25 30 35 40 45 50

```

Task 6: Write a C++ program that prints the sum of the first 10 natural numbers.

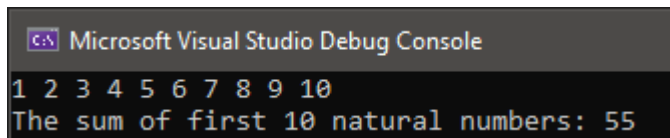
Code

```

#include <iostream>
using namespace std;
int main()
{
    int i, sum = 0;
    for (i = 1; i <= 10; i++){
        cout << i << " ";
        sum = sum + i;
    }
    cout << "\nThe sum of first 10 natural numbers: " << sum << endl;
}

```

Output



```

Microsoft Visual Studio Debug Console
1 2 3 4 5 6 7 8 9 10
The sum of first 10 natural numbers: 55

```

Task 7: Write a C++ program that prints the series 1 2 4 8 16 32 64 128.

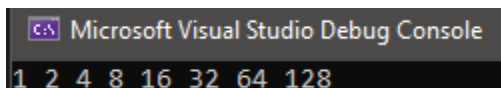
Code

```

#include <iostream>
using namespace std;
int main()
{
    for (int i = 1; i <= 128; i*=2){
        cout << i << " ";
    }
}

```

Output



```

Microsoft Visual Studio Debug Console
1 2 4 8 16 32 64 128

```

Lab Session 7 (Post-Lab Tasks)

Task 1: Write a C++ program that takes starting and ending value from user and prints only even numbers from start to end values.

Code

```

#include <iostream>
using namespace std;
int main()
{
    int a, b;
    char c;
    do{
        cout << "Please enter start value = ";
        cin >> a;
        cout << "Please enter end value = ";
    }
}

```

```

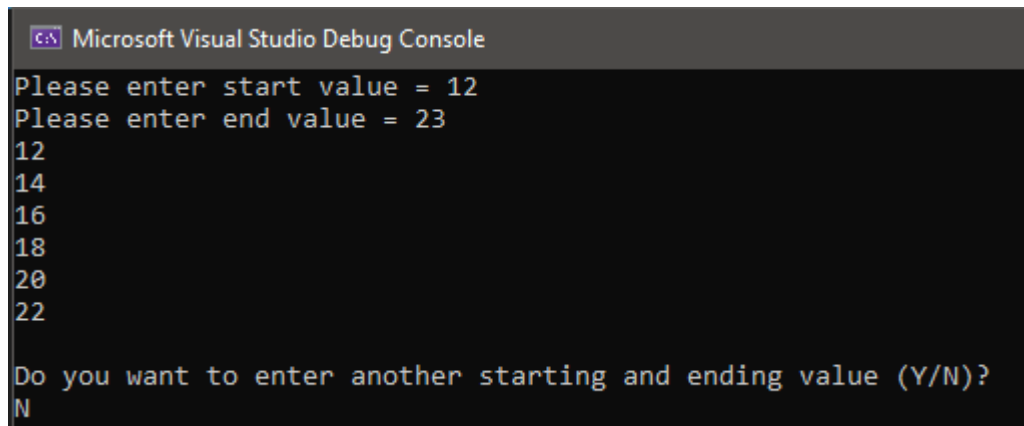
cin >> b;

for (int i = a; i <= b; i++) {
    if (i % 2 == 0) {
        cout << i << endl;
    }
}

cout << "\nDo you want to enter another starting and ending value (Y/N)? " << endl;
cin >> c;
} while (c == 'Y');
}

```

Output



```

Microsoft Visual Studio Debug Console
Please enter start value = 12
Please enter end value = 23
12
14
16
18
20
22
Do you want to enter another starting and ending value (Y/N)?
N

```

Task 2: Write a C++ program that takes input 'N' from user and print previous 10 odd numbers.

Task 3: Write a C++ program which take a number from user and print its factorials by using while loop.

Code

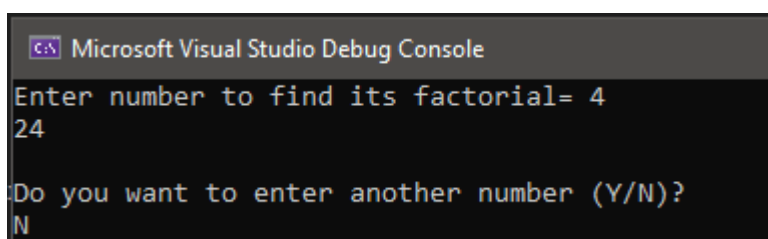
```

#include <iostream>
using namespace std;
int main()
{
    int N, factorial = 1;
    char c;
    do{
        cout << "Enter number to find its factorial= ";
        cin >> N;

        for (int i = 1; i <= N; i++) {
            factorial *= i;
        }
        cout << factorial << endl;
        cout << "\nDo you want to enter another number (Y/N)? " << endl;
        cin >> c;
    } while (c == 'Y');
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter number to find its factorial= 4
24
Do you want to enter another number (Y/N)?
N

```

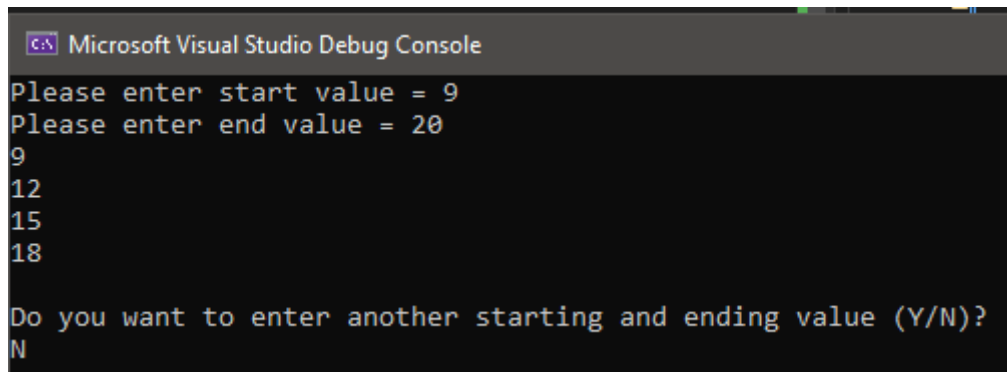
Task 4: Write a C++ program which will take two numbers from user, say start and end. Print the numbers from start to end that are divisible by 3.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int a, b;
    char c;
    do {
        cout << "Please enter start value = ";
        cin >> a;
        cout << "Please enter end value = ";
        cin >> b;

        for (int i = a; i <= b; i++) {
            if (i % 3 == 0) {
                cout << i << endl;
            }
        }
        cout << "\nDo you want to enter another starting and ending value (Y/N)? " << endl;
        cin >> c;
    } while (c == 'Y');
}
```

Output

The screenshot shows the Microsoft Visual Studio Debug Console with a dark background. The text is as follows:
Please enter start value = 9
Please enter end value = 20
9
12
15
18

Do you want to enter another starting and ending value (Y/N)?
N

Task 5: Write a C++ program to find whether a given number is odd or even. The program should continue as long as the user wants. Ask user to press 'e' or 'E' to exit the program.

Code

```
#include <iostream>
using namespace std;
int main()
{
    char c;
    do {
        int i;
        cout << "Enter the value: ";
        cin >> i;

        if (i % 2 == 1) {
            cout << i << " is an odd number" << endl;
        }
        else {
            cout << i << " is an even number" << endl;
        }

        cout << "\nPress 'e' or 'E' to exit the program" << endl;
        cin >> c;
    } while (c != 'e' && c != 'E');
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter the value: 13
13 is an odd number
Press 'e' or 'E' to exit the program
a
Enter the value: 15
15 is an odd number
Press 'e' or 'E' to exit the program
e
```

```
Microsoft Visual Studio Debug Console
Enter the value: 12
12 is an even number
Press 'e' or 'E' to exit the program
E
```

Task 6: Write a program that checks whether a number is prime or not by using if else structure and while loop.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int num1, ctr = 0;
    cout << "Input a number to check prime or not: ";
    cin >> num1;
    int a = 1;
    while (a <= num1){
        if (num1 % a == 0){
            ctr++;
        }
        a++;
    }
    if (ctr == 2){
        cout << "The entered number is a prime number\n";
    }
    else {
        cout << "The number you entered is not a prime number \n";
    }
}
```

Output

```
Microsoft Visual Studio Debug Console
Input a number to check prime or not: 4
The number you entered is not a prime number
```

```
Microsoft Visual Studio Debug Console
Input a number to check prime or not: 13
The entered number is a prime number
```

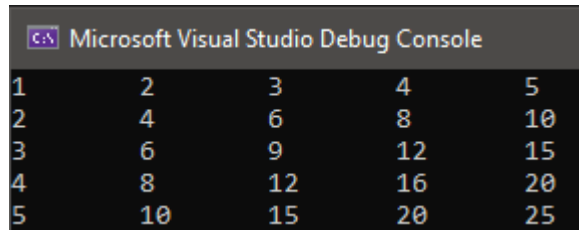
Lab Session 8 (Nested loops Tasks)

Task 1:

Code

```
#include <iostream>
using namespace std;
int main(){
    for (int r = 1; r <= 5; r++) {
        for (int c = 1; c <= 5; c++) {
            cout << r * c << "\t";
        }
        cout << endl;
    }
}
```

Output



1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

Task 2: what is the output of the following program?

```
#include<iostream>
using namespace std;
int main()
{
    int sum = 0;
    for (int i = 2; i <= 5; i++)
    {
        for (int j = 7; j <= 9; j++)
        {
            sum += (i * j);
        }
    }
    cout << sum;
    return 0;
}
```

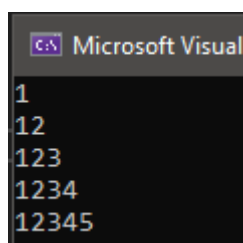
Output = 336

Task 3:

Code

```
#include <iostream>
using namespace std;
int main() {
    for (int r = 1; r <= 5; r++) {
        for (int c = 1; c <= r; c++) {
            cout << c;
        }
        cout << endl;
    }
}
```

Output



```
1
12
123
1234
12345
```

Task 4: Print factorial of all numbers from 1 to n. Where n is entered by the user.

Task 5:

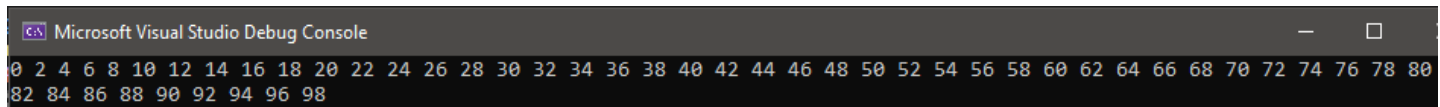
Practice Problems (simple loops)

Task 1: Write a program that prints the first 50 even numbers by using for loop.

Code

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 0; i < 100; i += 2) {
        cout << i << " ";
    }
}
```

Output

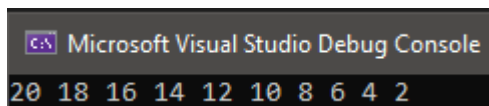


Task 2: Write a C++ program which will print even numbers from 20 to 1 by using for loop.

Code

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 20; i >= 1; i -= 2) {
        cout << i << " ";
    }
}
```

Output



Task 3: Write a C++ program which will take ten numbers from user and for each number display whether the given number is even or odd.

Code

```
#include <iostream>
using namespace std;
int main(){
    int i;
    int X = 1;
    while (X<= 10) {
        cout <<"\nEnter the value : ";
        cin >> i;
        if (i % 2 == 1) {
            cout << i << " is an odd number" << endl;
        }
        else
            cout << i << " is an even number" << endl;
        X++;
    }
}
```

Output


```
C:\> Select Microsoft Visual Studio Debug Console

Enter the value : 12
12 is an even number

Enter the value : 13
13 is an odd number

Enter the value : 88
88 is an even number

Enter the value : 78
78 is an even number

Enter the value : 65
65 is an odd number

Enter the value : 78
78 is an even number

Enter the value : 90
90 is an even number

Enter the value : 34
34 is an even number

Enter the value : 55
55 is an odd number

Enter the value : 66
66 is an even number
```

Task 4: Write a C++ program which will print first 10 numbers which are multiple of 7.

Code

```
#include <iostream>
using namespace std;
int main() {
    for (int i =7; i <= 70;i+=7){
        cout << i << " ";
    }
}
```

Output

```
C:\> Microsoft Visual Studio Debug Console
7 14 21 28 35 42 49 56 63 70
```

Task 5: Write a C++ program that prints the sum of the first 10 natural numbers.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int i, sum = 0;
    for (i = 1; i <= 10; i++){
        cout << i << " ";
        sum = sum + i;
    }
    cout << "\nThe sum of first 10 natural numbers: " << sum << endl;
}
```

Output

```
C:\N Microsoft Visual Studio Debug Console
1 2 3 4 5 6 7 8 9 10
The sum of first 10 natural numbers: 55
```

Task 6: Write a C++ program that prints the series 1 2 4 8 16 32 64 128.

Code

```
#include <iostream>
using namespace std;
int main()
{
    for (int i = 1; i <= 128; i*=2){
        cout << i << " ";
    }
}
```

Output

```
C:\N Microsoft Visual Studio Debug Console
1 2 4 8 16 32 64 128
```

Task 7: Write a C++ program that prints the series 3 6 9 12 15 18 21 24.....60

Code

```
#include <iostream>
using namespace std;
int main()
{
    for (int i = 1; i <= 20; i++){
        cout << i*3<<" ";
    }
}
```

Output

```
C:\N Microsoft Visual Studio Debug Console
3 6 9 12 15 18 21 24 27 30 33 36 39 42 45 48 51 54 57 60
```

Task 8: What will be the output of following code?

```
#include <iostream>
using namespace std;
int main()
{
    int x = 0, y = 0, a = 0, b = 7;
    while (a < b)
    {
        x = a + b;
        y = b - a;
        cout << x << " " << y << endl;
        if (a % 2 == 0 && b % 2 == 0)
        {
            a++;
        }
        else if (a % 2 == 1 && b % 2 == 1)
        {
            b--;
        }
        else
        {
            a++;
            b--;
        }
    }
    return 0;
}
```

Output

7 7

7 5

7 3

7 1

Task 9: Write a C++ program which will take numbers from user and add them until user enter 0 by using do-while loop.

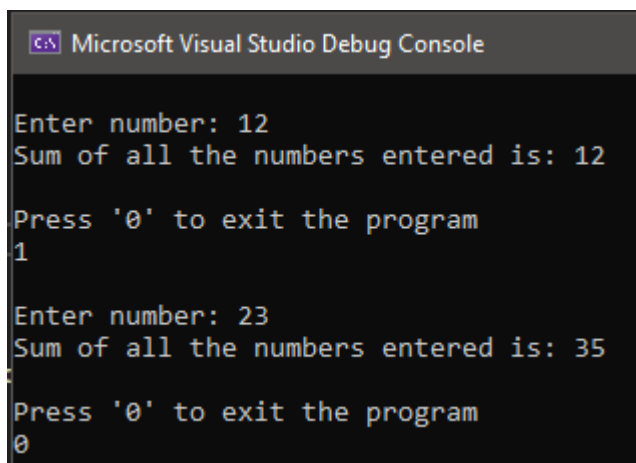
Code

```
#include <iostream>
using namespace std;
int main()
{
    int c, i, sum = 0;
    do {
        cout << "\nEnter number: ";
        cin >> i;

        sum = sum + i;
        cout << "Sum of all the numbers entered is: " << sum << endl;

        cout << "\nPress '0' to exit the program" << endl;
        cin >> c;
    } while (c != 0);
}
```

Output



```
Microsoft Visual Studio Debug Console

Enter number: 12
Sum of all the numbers entered is: 12

Press '0' to exit the program
1

Enter number: 23
Sum of all the numbers entered is: 35

Press '0' to exit the program
0
```

Task - 10: Write a do-while loop that asks the user to enter two numbers. The numbers should be added and the sum displayed. The user should be asked if he or she wishes to perform the operation again. If so, the loop should repeat; otherwise it should terminate – Sentinel Loop.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int c, i, sum, a;
    do {
        cout << "\nEnter first number: ";
```

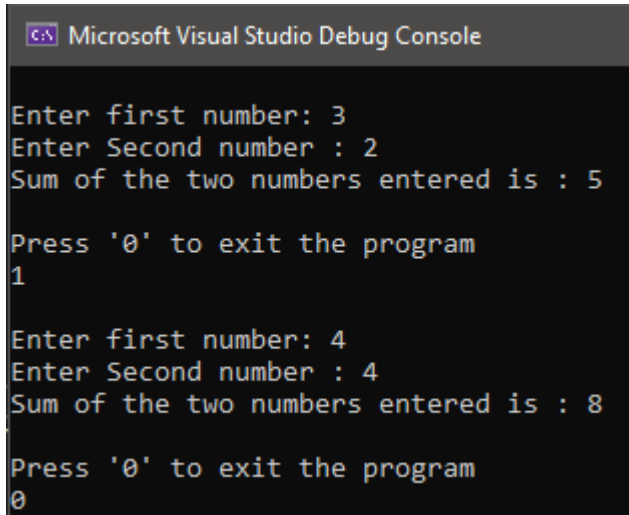
```

    cin >> i;
    cout << "Enter Second number : ";
    cin >> a;

    sum = a + i;
    cout << "Sum of the two numbers entered is : " << sum << endl;
    cout << "\nPress '0' to exit the program" << endl;
    cin >> c;
} while (c != 0);
}

```

Output



```

Microsoft Visual Studio Debug Console

Enter first number: 3
Enter Second number : 2
Sum of the two numbers entered is : 5

Press '0' to exit the program
1

Enter first number: 4
Enter Second number : 4
Sum of the two numbers entered is : 8

Press '0' to exit the program
0

```

Task - 11:

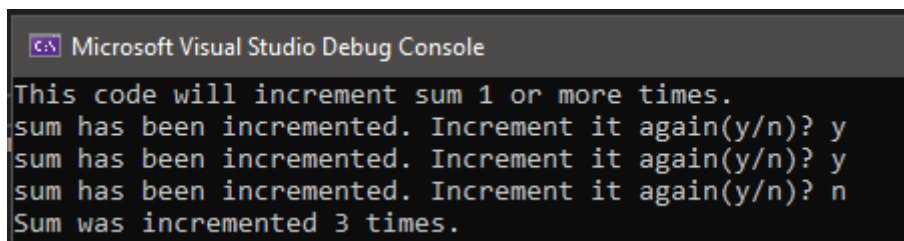
```

#include <iostream>
using namespace std;
int main()
{
    int sum = 0;
    char doAgain = 'y';
    cout << "This code will increment sum 1 or more times.\n";

    do {
        sum++;
        cout << "sum has been incremented. Increment it again(y/n)? ";
        cin >> doAgain;
    } while ((doAgain == 'y') || (doAgain == 'Y'));

    cout << "Sum was incremented " << sum << " times.\n";
}

```



```

Microsoft Visual Studio Debug Console

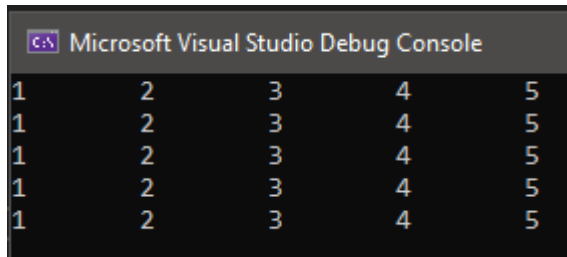
This code will increment sum 1 or more times.
sum has been incremented. Increment it again(y/n)? y
sum has been incremented. Increment it again(y/n)? y
sum has been incremented. Increment it again(y/n)? n
Sum was incremented 3 times.

```

Practice Problems (nested loops)

Task 1:

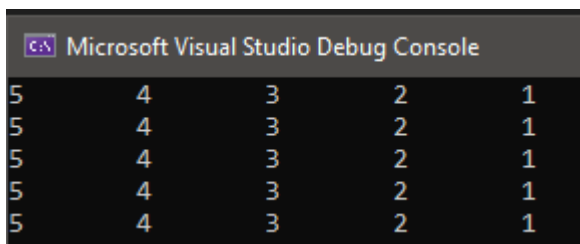
```
#include <iostream>
using namespace std;
int main() {
    for (int r = 1; r <= 5; r++) {
        for (int c = 1; c <= 5; c++) {
            cout << c<<"\t";
        }
        cout << endl;
    }
}
```



1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5

Task 2:

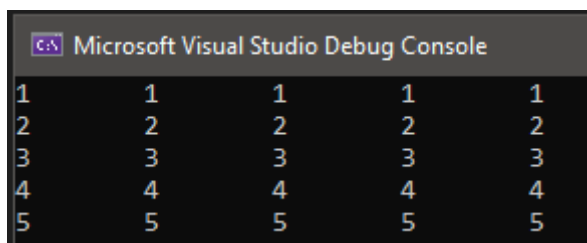
```
#include <iostream>
using namespace std;
int main() {
    for (int r = 1; r <= 5; r++) {
        for (int c = 5; c >= 1; c--) {
            cout << c<<"\t";
        }
        cout << endl;
    }
}
```



5	4	3	2	1
5	4	3	2	1
5	4	3	2	1
5	4	3	2	1
5	4	3	2	1

Task 3

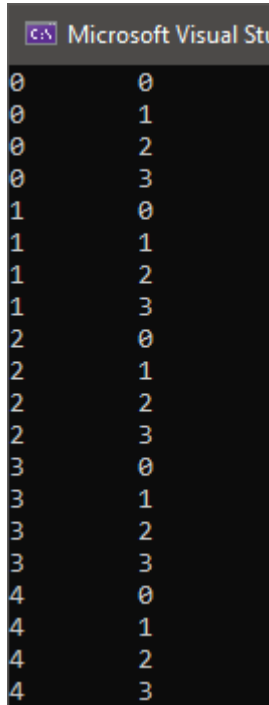
```
#include <iostream>
using namespace std;
int main() {
    for (int r = 1; r <= 5; r++) {
        for (int c = 5; c >= 1; c--) {
            cout << r<< "\t";
        }
        cout << endl;
    }
}
```



1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5

Task 4:

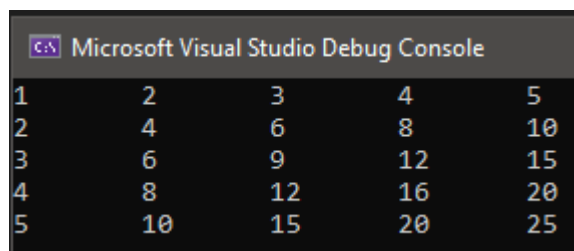
```
#include <iostream>
using namespace std;
int main() {
    for (int r = 0; r <= 4; r++) {
        for (int c = 0; c <= 3; c++) {
            cout << r << "\t" << c << endl;
        }
    }
}
```



```
C:\> Microsoft Visual Studio
0      0
0      1
0      2
0      3
1      0
1      1
1      2
1      3
2      0
2      1
2      2
2      3
3      0
3      1
3      2
3      3
4      0
4      1
4      2
4      3
```

Task 5:

```
#include <iostream>
using namespace std;
int main() {
    for (int r = 1; r <= 5; r++) {
        for (int c = 1; c <= 5; c++) {
            cout << r * c << "\t";
        }
        cout << endl;
    }
}
```



```
C:\> Microsoft Visual Studio Debug Console
1      2      3      4      5
2      4      6      8      10
3      6      9      12     15
4      8      12     16     20
5      10     15     20     25
```

Task 6:

```
#include<iostream>
using namespace std;
int main()
{
    int sum = 0;
```

```

for (int i = 2; i <= 5; i++)
{
    for (int j = 7; j <= 9; j++)
    {
        sum += (i * j);
    }
}
cout << sum;
return 0;
}

```

Output

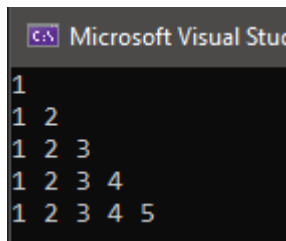
336

Task 7:

```

#include <iostream>
using namespace std;
int main() {
    for (int r = 1; r <= 5; r++) {
        for (int c = 1; c <= r; c++) {
            cout << c << " ";
        }
        cout << endl;
    }
}

```



```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

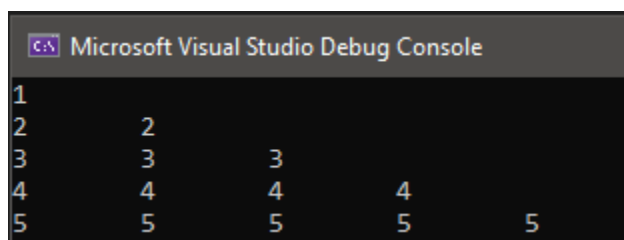
```

Task 8:

```

#include <iostream>
using namespace std;
int main() {
    for (int r = 1; r <= 5; r++) {
        for (int c = 1; c <= r; c++) {
            cout << r << "\t";
        }
        cout << endl;
    }
}

```



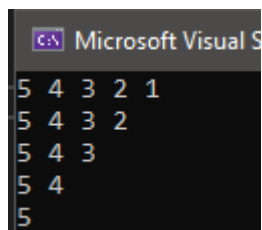
```

1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5

```

Task 9:

```
#include <iostream>
using namespace std;
int main() {
    for (int r = 1; r <= 5; r++) {
        for (int c = 5; c >= r; c--) {
            cout << c << " ";
        }
        cout << endl;
    }
}
```



Lab Session 8 (Functions Tasks)

Task 1: Using function, write a complete program that prints your name 10 times. The Function can take no arguments and should not return any value.

Code

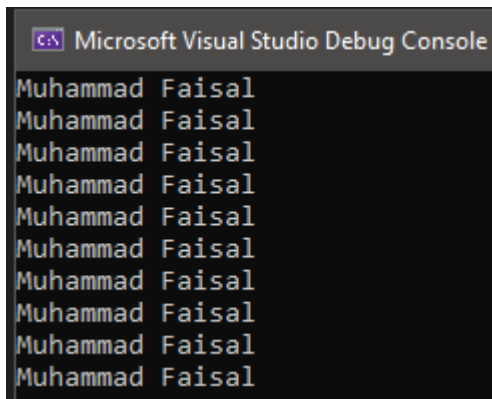
```
#include <iostream>
using namespace std;

void DisplayName();

int main() {
    DisplayName();
}

void DisplayName() {
    for (int i = 1; i <= 10; i++) {
        cout << "Muhammad Faisal" << endl;
    }
}
```

Output



Task 2: Write a function to calculate the factorial value of any integer entered through the keyboard.

Code

```
#include <iostream>
using namespace std;

int factorial(int);

int main() {
    int x;
    cout << "Enter value to find the factorial value: ";
    cin >> x;
    cout << "The factorial value of " << x << " is " << factorial(x) << endl;
    return 0;
}

int factorial(int a) {
    int fact = 1;
    for (int i = 1; i <= a; i++) {
        fact = fact * i;
    }
    return fact;
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter value to find the factorial value: 4
The factorial value of 4 is 24
```

Task 3: Write a function which receives a float and an int from main (), finds the product of these two and returns the product which is printed through main ().

Code

```
#include <iostream>
using namespace std;

float Product(float,int);

int main() {
    float x;
    int y;
    cout << "Enter one float and one integer value to find its product:\n";
    cin >> x >> y;
    cout << "The product is " << Product(x,y) << endl;
    return 0;
}

float Product(float a,int b) {
    float c;
    c = a * b;
    return c;
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter one float and one integer value to find its product:
2.3
12
The product is 27.6
```

Task 4: Given three variables x, y, z write a function to circularly shift their values to right. In other words if x = 5, y = 8, z = 10 after circular shift y = 5, z = 8, x = 10 after circular shift y = 5, z = 8 and x = 10. Call the function with variables a, b, c to circularly shift values.

Code

```
#include <iostream>
using namespace std;

void circular_shift(int,int,int);

int main() {
    int x,y,z;
    cout << "Enter X,Y and Z values to circularly shift them:\n";
    cin >> x >> y >> z;
    cout<<"Before circular shift: "<<"X = "<<x<<" Y = "<<y<<" Z = "<<z<<endl;
    circular_shift(x, y, z);
    return 0;
}

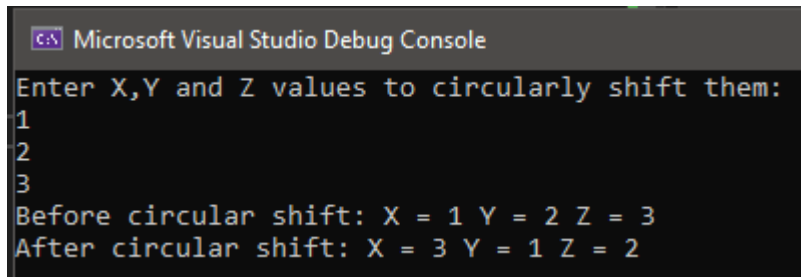
void circular_shift(int a, int b, int c){
    int temp;
    temp = c;
```

```

c = b;
b = a;
a = temp;
cout<<"After circular shift: "<< "X = "<<a<< " Y = "<<b<< " Z = "<<c<< endl;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter X,Y and Z values to circularly shift them:
1
2
3
Before circular shift: X = 1 Y = 2 Z = 3
After circular shift: X = 3 Y = 1 Z = 2

```

Task 5:

Code

```

#include <iostream>
#include <cmath>
using namespace std;

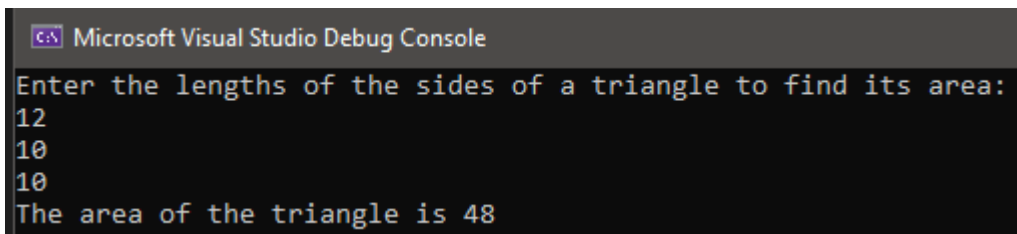
int area(int,int,int);

int main() {
    int d,e,f;
    cout << "Enter the lengths of the sides of a triangle to find its
area:\n";
    cin >> d >> e >> f;
    cout << "The area of the triangle is " << area(d,e,f) << endl;
    return 0;
}

int area(int a, int b, int c){
    int x,s;
    s = (a + b + c) / 2;
    x = sqrt(s*(s - a)*(s - b)*(s - c));
    return x;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter the lengths of the sides of a triangle to find its area:
12
10
10
The area of the triangle is 48

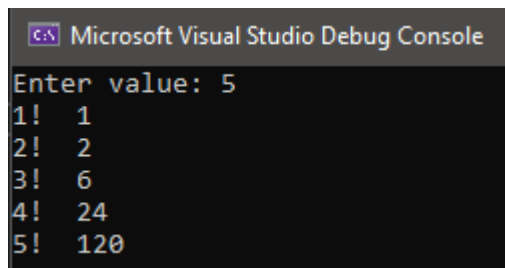
```

Task 4: Print factorial of all numbers from 1 to n. Where n is entered by the user.

Code

```
#include <iostream>
using namespace std;
int main() {
    int n, factorial=1;
    cout << "Enter value: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        factorial = factorial * i;
        cout << i << "! " << factorial << endl;
    }
    return 0;
}
```

Output



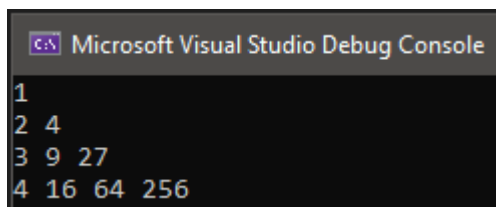
```
Microsoft Visual Studio Debug Console
Enter value: 5
1! 1
2! 2
3! 6
4! 24
5! 120
```

Task 5:

Code

```
#include <iostream>
#include <math.h>
using namespace std;
int main() {
    for (int i = 1; i <= 4; i++) {
        for (int x = 1; x <= i; x++) {
            cout << pow(i,x) << " ";
        }
        cout << endl;
    }
}
```

Output



```
Microsoft Visual Studio Debug Console
1
2 4
3 9 27
4 16 64 256
```

Lab session 9(In lab task)

Task 1: Write a C++ program to find the largest element of a given array of integers.

Code

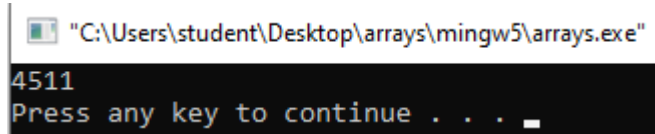
```
#include<iostream>
using namespace std;

int main()
{
    int array[10] = {
        12, 4511, 26, 8, 780, 112, 49, 36, 79, 351
    } ;
    int highest = array[0];

    for(int i=1; i<10; i++)
    {
        if(array[i] > highest)
        {
            highest = array[i];
        }
    }

    cout<<highest<<endl;
}
```

Output



```
"C:\Users\student\Desktop\arrays\mingw5\arrays.exe"
4511
Press any key to continue . . . _
```

Task 2: Write a C++ program to count the total number of positive, negative and zeros in a given array.

Code

```
#include<iostream>
using namespace std;

int main()
{
    int array[10] = {12, -2, -5, 0, 78, 11, 0, 0, -10, -9};
    int n=0,z=0,p=0;

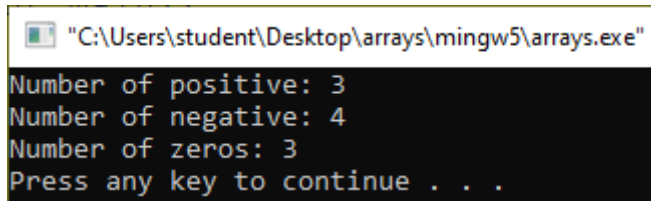
    for(int i=0; i<10; i++)
    {
        if(array[i] > 0)
        {
            p=p+1;
        }
        else if(array[i] < 0){
            n=n+1;
        }
        else{
            z=z+1;
        }
    }
}
```

```

    }
    cout<<"Number of positive: "<<p<<endl;
    cout<<"Number of negative: "<<n<<endl;
    cout<<"Number of zeros: "<<z<<endl;
}

```

Output



```

"C:\Users\student\Desktop\arrays\mingw5\arrays.exe"
Number of positive: 3
Number of negative: 4
Number of zeros: 3
Press any key to continue . . .

```

Task 3: Write a program to print elements of an array in reverse order.

Code

```

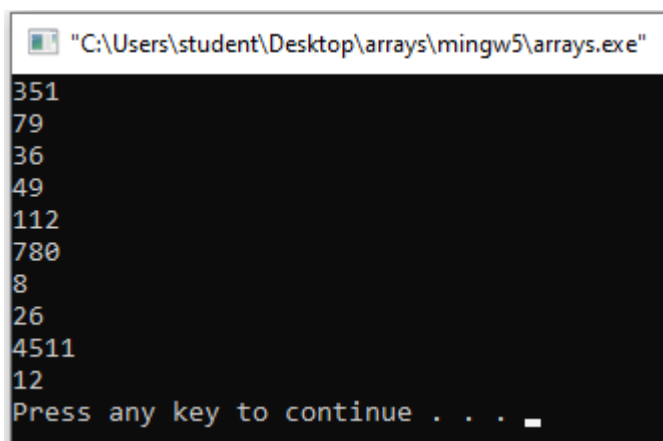
#include<iostream>
using namespace std;

int main()
{
    int array[10] = {
        12, 4511, 26, 8, 780, 112, 49, 36, 79, 351
    };
    int highest = array[0];

    for(int i=9; i>=0; i--)
    {
        cout<<array[i]<<endl;
    }
}

```

Output



```

"C:\Users\student\Desktop\arrays\mingw5\arrays.exe"
351
79
36
49
112
780
8
26
4511
12
Press any key to continue . . .

```

Task 4: Write a program that accepts temperature of 7 days from user and print the weekly average temperature using array.

Code

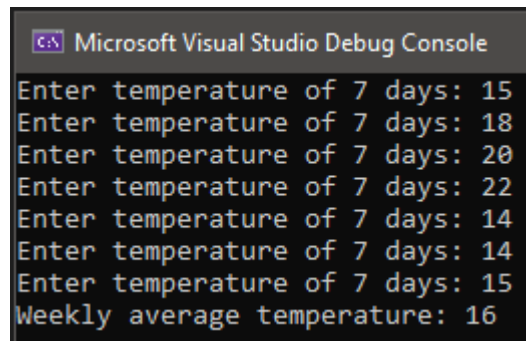
```
#include<iostream>
using namespace std;

int main()
{
    int temp[7];
    int total = 0;
    int avg;

    for (int i = 0; i < 7; i++)
    {
        cout << "Enter temperature of 7 days: ";
        cin >> temp[i];
        total = total + temp[i];
        avg = (total / 7);
    }

    cout << "Weekly average temperature: " << avg << endl;
}
```

Output



The screenshot shows the Microsoft Visual Studio Debug Console with the following output:

```
Enter temperature of 7 days: 15
Enter temperature of 7 days: 18
Enter temperature of 7 days: 20
Enter temperature of 7 days: 22
Enter temperature of 7 days: 14
Enter temperature of 7 days: 14
Enter temperature of 7 days: 15
Weekly average temperature: 16
```

Lab session 9(Post-lab task)

Task 1: Write a C++ program to update every array element by multiplication of next and previous values of a given array of integers.

Code

```
#include<iostream>
using namespace std;

void replace_elements(int nums[], int n);

int main()
{
    int nums[] = { 0, 1, 3, 4, 5, 6, 7, 8, 10 };
    int n = sizeof(nums) / sizeof(nums[0]);
    cout << "Original array: ";
    for (int i = 0; i < n; i++)
        cout << nums[i] << " ";
    replace_elements(nums, n);

    cout << "\nNew array elements: ";
    for (int i = 0; i < n; i++)
        cout << nums[i] << " ";

    return 0;
}

void replace_elements(int nums[], int n)
{
    if (n <= 1)
        return;

    int previous_element = nums[0];
    nums[0] = nums[0] * nums[1];

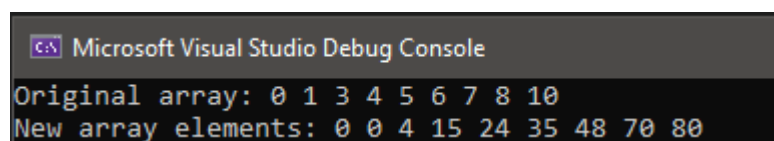
    for (int i = 1; i < n - 1; i++)
    {
        int current_element = nums[i];

        nums[i] = previous_element * nums[i + 1];

        previous_element = current_element;
    }

    nums[n - 1] = previous_element * nums[n - 1];
}
```

Output



```
Microsoft Visual Studio Debug Console
Original array: 0 1 3 4 5 6 7 8 10
New array elements: 0 0 4 15 24 35 48 70 80
```


Task 2. Write a C++ program to separate even and odd numbers of an array of integers. Put all even numbers first, and then odd numbers.

Code

```
#include<iostream>
using namespace std;

void swap(int* x, int* y);

void separate_evenodd(int nums[], int size);

int main()
{
    int nums[] = { 0, 1, 3, 4, 5, 6, 7, 8, 10 };
    int n = sizeof(nums) / sizeof(nums[0]);
    cout << "Original array: ";
    for (int i = 0; i < n; i++)
        cout << nums[i] << " ";
    separate_evenodd(nums, n);

    cout<<"\nArray after divided: ";
    for (int i = 0; i < n; i++)
        cout << nums[i] << " ";
    return 0;
}

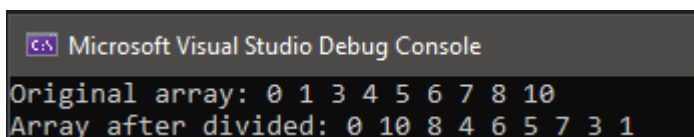
void swap(int* x, int* y)
{
    int temp = *x;
    *x = *y;
    *y = temp;
}

void separate_evenodd(int nums[], int size)
{
    int left_num = 0, right_num = size - 1;
    while (left_num < right_num)
    {
        while (nums[left_num] % 2 == 0 && left_num < right_num)
            left_num++;

        while (nums[right_num] % 2 == 1 && left_num < right_num)
            right_num--;

        if (left_num < right_num)
        {
            swap(&nums[left_num], &nums[right_num]);
            left_num++;
            right_num--;
        }
    }
}
```

Output



```
C:\> Microsoft Visual Studio Debug Console
Original array: 0 1 3 4 5 6 7 8 10
Array after divided: 0 10 8 4 6 5 7 3 1
```

Task 3: Write a C++ program to find second largest element in a given array of integers.

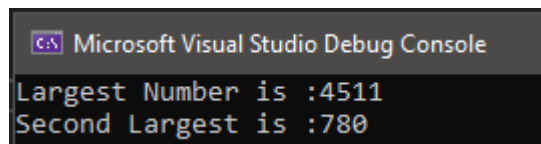
Code

```
#include<iostream>
using namespace std;

int main()
{
    int array[10] = {
        12, 4511, 26, 8, 780, 112, 49, 36, 79, 351
    };
    int largest = array[1];
    int second_largest = array[2];

    if (largest < second_largest)
    {
        largest = array[2];
        second_largest = array[1];
    }
    for (int i = 2; i < 10; i++)
    {
        if (array[i] > largest)
        {
            second_largest = largest;
            largest = array[i];
        }
        else if (array[i] > second_largest)
        {
            second_largest = array[i];
        }
    }
    cout << "Largest Number is :" << largest<< endl;
    cout << "Second Largest is :" << second_largest<< endl;
}
```

Output



```
C:\> Microsoft Visual Studio Debug Console
Largest Number is :4511
Second Largest is :780
```

Task 4: Write a C++ program to find the most occurring element in an array of integers.

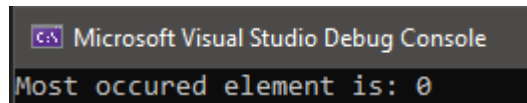
Code

```
#include <iostream>
using namespace std;
int main()
{
    int i, j;
    int num[] = { 0, 0, 1, 1, 0, 0, 7, 8, 10 };
    int popular = num[0];
    int temp = 0, tempCount, count = 1;

    for (i = 0; i < 5; i++)
    {
        tempCount = 0;
        temp = num[i];
        tempCount++;
    }
```

```
for (j = i + 1; j < 5; j++)
{
    if (num[j] == temp)
    {
        tempCount++;
        if (tempCount > count)
        {
            popular = temp;
            count = tempCount;
        }
    }
}
cout << "Most occured element is: " << popular;
}
```

Output



Microsoft Visual Studio Debug Console
Most occured element is: 0

Task 2: Write a C++ program that takes input 'N' from user and print previous 10 odd numbers.

Code

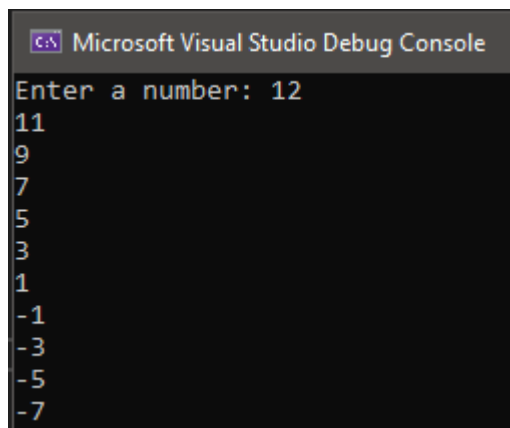
```
#include<iostream>
using namespace std;

int main()
{
    int n;

    cout << "Enter a number: ";
    cin >> n;

    for (int i = n-1; i > n-20; i--)
    {
        if (i % 2 != 0) {
            cout << i << endl;
        }
    }
}
```

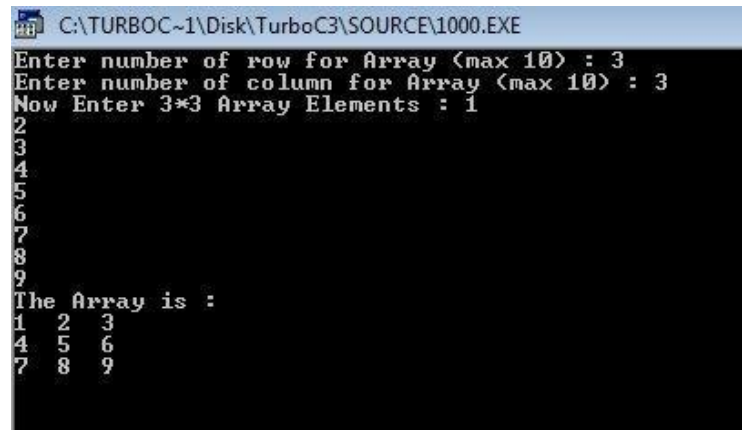
Output

A screenshot of the Microsoft Visual Studio Debug Console. The title bar reads "Microsoft Visual Studio Debug Console". The console output shows the prompt "Enter a number: 12" followed by a list of 10 odd numbers: 11, 9, 7, 5, 3, 1, -1, -3, -5, and -7, each on a new line.

```
Microsoft Visual Studio Debug Console
Enter a number: 12
11
9
7
5
3
1
-1
-3
-5
-7
```

Lab session 10(In-lab task)

Task 1: When the above C++ program is compile and executed, it will produce the following result:



```
C:\TURBOC~1\Disk\TurboC3\SOURCE\1000.EXE
Enter number of row for Array <max 10> : 3
Enter number of column for Array <max 10> : 3
Now Enter 3*3 Array Elements : 1
2
3
4
5
6
7
8
9
The Array is :
1 2 3
4 5 6
7 8 9
```

Code

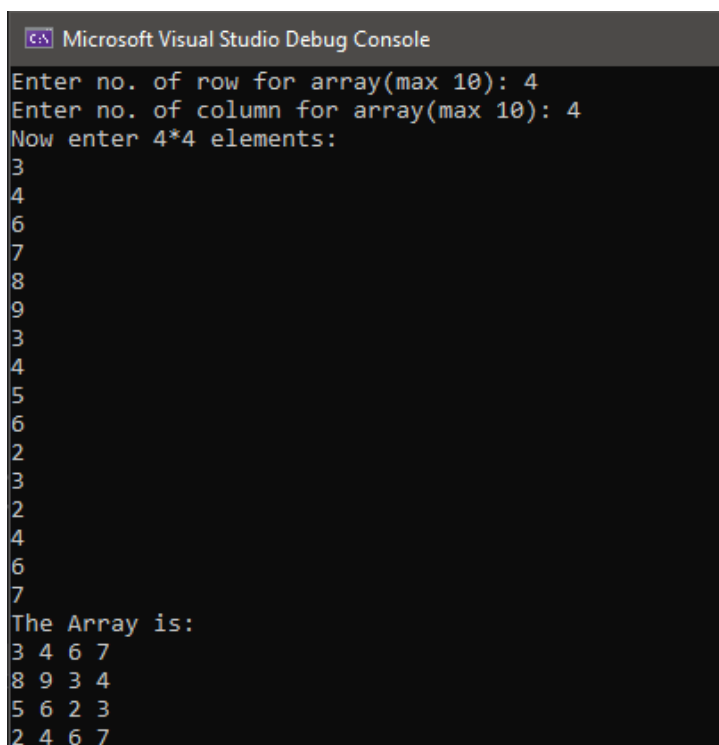
```
#include <iostream>
using namespace std;

int main() {
    int r, c;
    int array[10][10];

    cout << "Enter no. of row for array(max 10): ";
    cin >> r;
    cout << "Enter no. of column for array(max 10): ";
    cin >> c;
    array[r][c];

    cout << "Now enter " << r << "*" << c << " elements:" << endl;
    for (int i = 0; i < r; i++) {
        for (int y = 0; y < c; y++) {
            cin >> array[i][y];
        }
    }
    cout << "The Array is: " << endl;
    for (int i = 0; i < r; i++) {
        for (int y = 0; y < c; y++) {
            cout << array[i][y]<<" ";
        }
        cout << endl;
    }
}
```

Output



```
Microsoft Visual Studio Debug Console
Enter no. of row for array(max 10): 4
Enter no. of column for array(max 10): 4
Now enter 4*4 elements:
3
4
6
7
8
9
3
4
5
6
2
3
2
4
6
7
The Array is:
3 4 6 7
8 9 3 4
5 6 2 3
2 4 6 7
```

Task 2: Write a function named "sum" that takes as its arguments the following:

- (1) an array of floating point values;
- (2) an integer that tells how many floating point values are in the array.

The function should return as its value the sum of the floating point values in the array. Thus, for example, if the array that's passed to the function looks like this:

0	1	2	3	4
5.8	2.6	9.0	3.4	7.1

then the function should return the value 27.9 as its value.

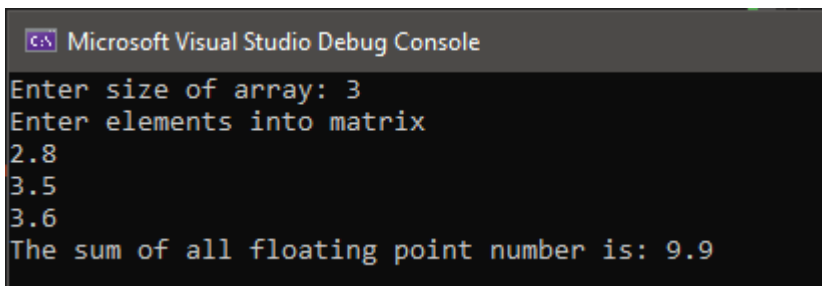
Code

```
#include <iostream>
using namespace std;
float sum(float a[], int x);

int main() {
    int y;
    cout << "Enter size of array: ";
    cin >> y;
    float* array = new float[y];
    cout << "Enter elements into matrix " << endl;
    for (int z = 0; z < y; z++) {
        cin >> array[z];
    }
    cout << "The sum of all floating point number is: " << sum(array, y) << endl;
}

float sum(float a[], int x) {
    float s = 0;
    for (int i = 0; i < x; i++) {
        s = s + a[i];
    }
    return s;
}
```

Output



```
Microsoft Visual Studio Debug Console
Enter size of array: 3
Enter elements into matrix
2.8
3.5
3.6
The sum of all floating point number is: 9.9
```

Task 3: Write a function named "location_of_largest" that takes as its arguments the following:

- (1) an array of integer values;
- (2) an integer that tells how many integer values are in the array.

The function should return as its value the subscript of the cell containing the largest of the values in the array.

Thus, for example, if the array that's passed to the function looks like this:

0	1	2	3	4
58	26	90	34	71

then the function should return the integer 2 as its value. If there is more than one cell containing the largest of the values in the array, then the function should return the *smallest* of the subscripts of the cells containing the largest values. For example, if the array that's passed to the function is

0	1	2	3	4	5	6
58	26	91	34	70	91	88

then the largest value occurs in cells 2 and 5, so the function should return the integer value 2.

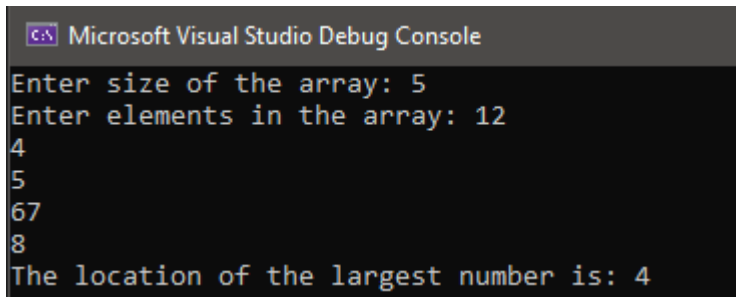
Code

```
#include <iostream>
using namespace std;
int location_of_largest(int a[], int x);

int main() {
    int y;
    cout << "Enter size of the array: ";
    cin >> y;
    int* array= new int[y];

    cout << "Enter elements in the array: ";
    for (int z = 0; z < y; z++) {
        cin>>array[z];
    }
    cout << "The location of the largest number is: " << location_of_largest(array, y) << endl;
}
int location_of_largest(int a[], int x) {
    int largest = a[0];
    for (int i = 1; i < x; i++) {
        if (largest < a[i]) {
            i = i + 1;
            return i;
        }
    }
}
```

Output



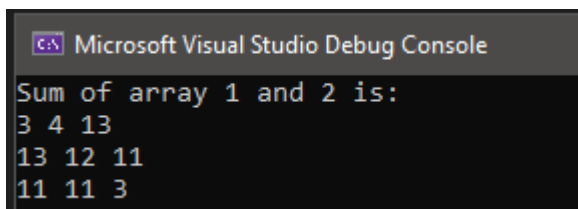
```
Microsoft Visual Studio Debug Console
Enter size of the array: 5
Enter elements in the array: 12
4
5
6
7
8
The location of the largest number is: 4
```

Task 4: Write a C++ program to add two 3x3 matrices by using multi-dimensional arrays.

Code

```
#include <iostream>
using namespace std;
void sum(int array1[3][3], int array2[3][3]);
int main() {
    int array1[3][3] = { 2,3,4,5,5,5,6,7,1 };
    int array2[3][3] = { 1,1,9,8,7,6,5,4,2 };
    sum(array1, array2);
}
void sum(int array1[3][3], int array2[3][3]) {
    cout << "Sum of array 1 and 2 is: " << endl;
    for (int a = 0; a < 3; a++) {
        for (int b = 0; b < 3; b++) {
            cout << array1[a][b] + array2[a][b] << " ";
        }
        cout << endl;
    }
}
```

Output



```
Microsoft Visual Studio Debug Console
Sum of array 1 and 2 is:
3 4 13
13 12 11
11 11 3
```

Lab session 10(Post-lab task)

Task 1:

- (1) an array of floating point values;
- (2) an integer that tells the number of cells in the array;

The function should shift the contents of each cell one place to the right, except for the contents of the last cell, which should be moved into the cell with subscript 0 . Thus, for example, if the array passed to the function looks like this:

0	1	2	3	4
5.8	2.6	9.1	3.4	7.0

then when the function returns, the array will have been changed so that it looks like this:

0	1	2	3	4
7.0	5.8	2.6	9.1	3.4

The function should not return a value.

Code

```
#include <iostream>
using namespace std;

void shift(float array[], int y);

int main() {
    int x;
    int n;
    cout << "Enter size of array: ";
    cin >> x;
    float* array = new float[x];

    cout << "Enter elements of array: " << endl;
    for (int i = 0; i < x; i++) {
        cin >> array[i];
    }
    shift(array, x);
    cout << "Array after shifting is ";
    for (int i = 0; i < x; i++) {
        cout << array[i] << " ";
    }
}

void shift(float array[], int y) {
    {
        float x = array[y - 1];
        for (int i = y - 1; i > 0; i--) {
            array[i] = array[i - 1];
        }
        array[0] = x;
    }
}
```

Output

```
C:\> Microsoft Visual Studio Debug Console
Enter size of array: 4
Enter elements of array:
2.3
3.4
5.5
4.6
Array after shifting is 4.6 2.3 3.4 5.5
```


Task 2: Write a menu driven C++ program to do following operation on two dimensional array A of size m x n. You should use user-defined functions which accept 2-D array A, and its size m and n as arguments. The options are:

- a. To input elements into matrix of size m x n
- b. To display elements of matrix of size m x n
- c. Sum of all elements of matrix of size m x n
- d. To display row-wise sum of matrix of size m x n
- e. To display column-wise sum of matrix of size m x n
- f. To create transpose of matrix B of size n x m

Code

```
#include <iostream>
using namespace std;

int main() {
    char z;
    int choice;
    int r, c;
    int A[10][10], B[10][10];
    cout << "Enter no. rows of array: ";
    cin >> r;
    cout << "Enter no. of columns of array: ";
    cin >> c;

    do {

        cout << "Choose from the following options: " << endl;
        cout << "1. To input elements into matrix of size m x n" << endl;
        cout << "2. To display elements of matrix of size m x n" << endl;
        cout << "3. Sum of all elements of matrix of size m x n" << endl;
        cout << "4. To display row-wise sum of matrix of size m x n" << endl;
        cout << "5. To display column-wise sum of matrix of size m x n" << endl;
        cout << "6. To create transpose of matrix B of size n x m" << endl;
        cin >> choice;

        switch (choice) {
            case 1:
                cout << "Enter elements into matrix " << endl;
                for (int i = 0; i < r; i++) {
                    for (int j = 0; j < c; j++) {
                        cin >> A[i][j];
                    }
                }
                break;

            case 2:
                cout << "Elements of matrix are: " << endl;
                for (int i = 0; i < r; i++) {
                    for (int j = 0; j < c; j++) {
                        cout << A[i][j] << " ";
                    }
                    cout << endl;
                }
                break;

            case 3:
                cout << "Sum of all elements of matrix is: ";

                for (int i = 0; i < r; i++) {
                    int sum2 = 0;
                    int x = 0;
                    for (int j = 0; j < c; j++) {
                        sum2 = sum2 + A[i][j];
                        x = sum2 + sum2;
                    }
                    cout << x << endl;
                }
                break;

            case 4:
                for (int i = 0; i < r; i++) {
                    int sum = 0;

```

```

        for (int j = 0; j < c; j++) {
            sum = sum + A[i][j];
        }
        cout << "Sum of elements of Row: " << i + 1 << " is " << sum << endl;
    }
    break;
case 5:
    for (int j = 0; j < c; j++) {
        int sum1 = 0;
        for (int i = 0; i < r; i++) {
            sum1 = sum1 + A[i][j];
        }
        cout << "Sum of elements of Column: " << j + 1 << " is " << sum1 <<
endl;
    }
    break;
case 6:
    for (int j = 0; j < c; j++) {
        for (int i = 0; i < r; i++) {
            B[j][i] = A[i][j];
        }
    }
    cout << "Transpose of matrix B of size n x m is " << endl;
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            cout << B[i][j] << " ";
        }
        cout << endl;
    }
    break;
}
cout << "Do you want to perform another operation (y/n)";
cin >> z;
}
while (z == 'y' || z == 'Y');
}

```

Output

C:\D:\sir syed\Programming fund practical\lab tasks\Lab Tasks\x64\Debug\Lab Tasks.exe

```

Enter no. rows of array: 2
Enter no. of columns of array: 2
Choose from the following options:
1. To input elements into matrix of size m x n
2. To display elements of matrix of size m x n
3. Sum of all elements of matrix of size m x n
4. To display row-wise sum of matrix of size m x n
5. To display column-wise sum of matrix of size m x n
6. To create transpose of matrix B of size n x m
1
Enter elements into matrix
3
2
4
1
Do you want to perform another operation (y/n) y
Choose from the following options:
1. To input elements into matrix of size m x n
2. To display elements of matrix of size m x n
3. Sum of all elements of matrix of size m x n
4. To display row-wise sum of matrix of size m x n
5. To display column-wise sum of matrix of size m x n
6. To create transpose of matrix B of size n x m
2
Elements of matrix are:
3 2
4 1
Do you want to perform another operation (y/n) y
Choose from the following options:
1. To input elements into matrix of size m x n
2. To display elements of matrix of size m x n
3. Sum of all elements of matrix of size m x n
4. To display row-wise sum of matrix of size m x n
5. To display column-wise sum of matrix of size m x n
6. To create transpose of matrix B of size n x m
3
Sum of all elements of matrix is: 10

```

```

Do you want to perform another operation (y/n) y
Choose from the following options:
1. To input elements into matrix of size m x n
2. To display elements of matrix of size m x n
3. Sum of all elements of matrix of size m x n
4. To display row-wise sum of matrix of size m x n
5. To display column-wise sum of matrix of size m x n
6. To create transpose of matrix B of size n x m
4
Sum of elements of Row: 1 is 5
Sum of elements of Row: 2 is 5
Do you want to perform another operation (y/n) y
Choose from the following options:
1. To input elements into matrix of size m x n
2. To display elements of matrix of size m x n
3. Sum of all elements of matrix of size m x n
4. To display row-wise sum of matrix of size m x n
5. To display column-wise sum of matrix of size m x n
6. To create transpose of matrix B of size n x m
5
Sum of elements of Column: 1 is 7
Sum of elements of Column: 2 is 3
Do you want to perform another operation (y/n) y
Choose from the following options:
1. To input elements into matrix of size m x n
2. To display elements of matrix of size m x n
3. Sum of all elements of matrix of size m x n
4. To display row-wise sum of matrix of size m x n
5. To display column-wise sum of matrix of size m x n
6. To create transpose of matrix B of size n x m
6
Transpose of matrix B of size n x m is
3 4
2 1

```

3. Write user-defined functions for a square matrix to calculate:
- Left diagonal sum
 - Right diagonal sum

Code

```

#include <iostream>
using namespace std;

int main() {
    int size;
    int A[10][10];
    int sum_left = 0, sum_right=0;
    cout << "Enter no. rows of array: ";
    cin >> size;
    cout << "Enter no. of columns of array: ";
    cin >> size;

    cout << "Enter elements into matrix " << endl;
    for (int i = 0; i < size; i++) {
        for (int j = 0; j < size; j++) {
            cin >> A[i][j];
            if (i == j)
                sum_left += A[i][j];
            if ((i + j) == size - 1)
                sum_right += A[i][j];
        }
    }
    cout << "Sum of Left Diagonal: " << sum_left << endl;
    cout << "Sum of Right Diagonal: " << sum_right << endl;
}

```

Output

```
Microsoft Visual Studio Debug Console
Enter no. rows of array: 3
Enter no. of columns of array: 3
Enter elements into matrix
1
2
3
4
5
6
7
5
4
Sum of Left Diagonal: 10
Sum of Right Diagonal: 15
```

Lab session 11

Task 1: Write a program to find a substring within a string. If found display its starting position.

Code

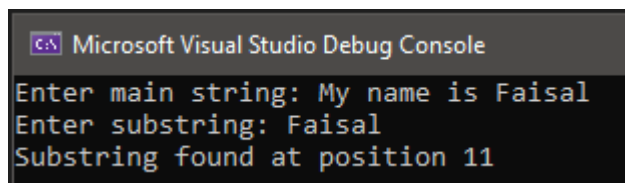
```
#include <iostream>
#include <string>
using namespace std;
int main()
{
    int pos;
    string mainstring, substring;

    cout << "Enter main string: ";
    getline(cin, mainstring);
    cout << "Enter substring: ";
    getline(cin, substring);

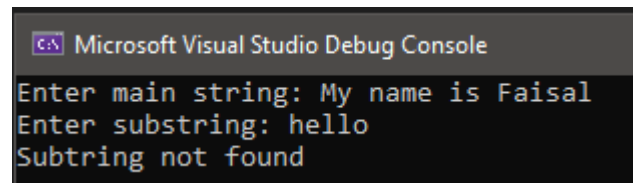
    pos = mainstring.find(substring);

    if (pos != string::npos) {
        cout << "Substring found at position " << pos << endl;
    }
    else {
        cout << "Subtring not found" << endl;
    }
    return 0;
}
```

Output



```
C# Microsoft Visual Studio Debug Console
Enter main string: My name is Faisal
Enter substring: Faisal
Substring found at position 11
```



```
C# Microsoft Visual Studio Debug Console
Enter main string: My name is Faisal
Enter substring: hello
Subtring not found
```

Task 2: Write program takes a string object from the user and calculates the number of vowels, digits and white-spaces.

Code

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
    int vowels = 0, spaces = 0, digits = 0;
    int len;
    string string;

    cout << "Enter a string: ";
    getline(cin, string);

    len = string.length();

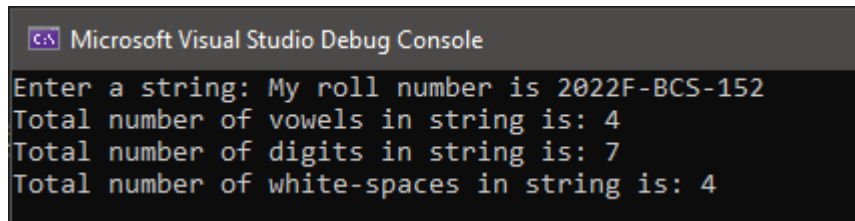
    for (int i = 0; i < len; i++) {
        if (string[i] == 'u' || string[i] == 'a' || string[i] == 'e' || string[i] == 'i' ||
string[i] == 'o' || string[i] == 'A' || string[i] == 'E' || string[i] == 'I' || string[i] == 'O'
|| string[i] == 'U') {
            vowels++;
        }
        else if (string[i] >= 48 && string[i] <= 57) {
            digits++;
        }
        else if (string[i] == ' ' || string[i] == '\t' || string[i] == '\n') {
            spaces++;
        }
    }
}
```

```

    }
    cout << "Total number of vowels in string is: " << vowels << endl;
    cout << "Total number of digits in string is: " << digits << endl;
    cout << "Total number of white-spaces in string is: " << spaces << endl;
    return 0;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter a string: My roll number is 2022F-BCS-152
Total number of vowels in string is: 4
Total number of digits in string is: 7
Total number of white-spaces in string is: 4

```

Task 3: Write a function to check whether a string “s” is a palindrome or not.

Code

```

#include <iostream>
#include <string>
using namespace std;

void isPalindrome(string s);

int main() {
    string input;
    cout << "Enter a string: ";
    getline(std::cin, input);

    isPalindrome(input);

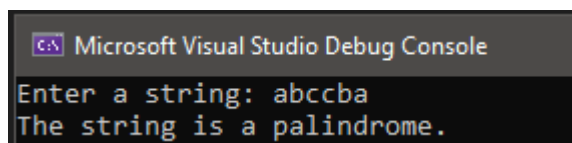
    return 0;
}

void isPalindrome(string s) {
    int left = 0;
    int right = s.length() - 1;

    while (left < right) {
        if (s[left] != s[right]) {
            cout << "The string is not a palindrome." << std::endl;
            return;
        }
        left++;
        right--;
    }
    cout << "The string is a palindrome." << std::endl;
}

```

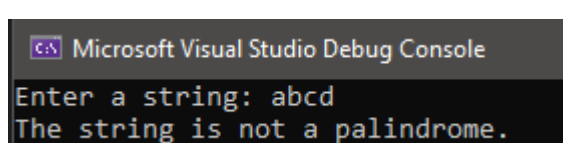
Output



```

Microsoft Visual Studio Debug Console
Enter a string: abccba
The string is a palindrome.

```



```

Microsoft Visual Studio Debug Console
Enter a string: abcd
The string is not a palindrome.

```

Task 4: Write a program to convert a string in lowercase.

Code

```

#include <iostream>
#include <string>
using namespace std;

int main()
{

```

```

int len;
string string;

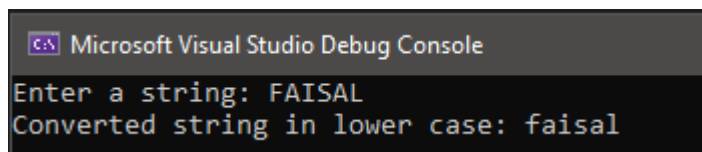
cout << "Enter a string: ";
getline(cin, string);

len = string.length();

for (int i = 0; i < len; i++) {
    if (string[i] >= 65 && string[i] <= 90) {
        string[i] = string[i] + 32;
    }
}
cout << "Converted string in lower case: " << string << endl;
return 0;
}

```

Output



```

C:\> Microsoft Visual Studio Debug Console
Enter a string: FAISAL
Converted string in lower case: faisal

```

Task 5: Write a C++ program to reverse a given string.

Code

```

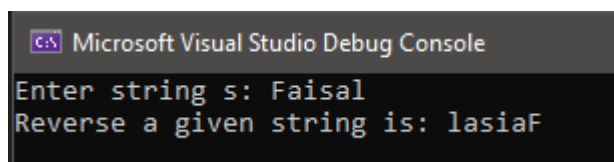
#include <iostream>
#include <string>
using namespace std;

int main()
{
    string s;
    cout << "Enter string s: ";
    getline(cin, s);

    reverse(s.begin(), s.end());
    cout << "Reverse a given string is: " << s << endl;
}

```

Output



```

C:\> Microsoft Visual Studio Debug Console
Enter string s: Faisal
Reverse a given string is: lasiaF

```

Task 6: Write a C++ program to capitalize the first letter of each word of a given string. Words must be separated by only one space.

Code

```

#include <iostream>
#include <string>
using namespace std;

int main() {
    string input;
    cout << "Enter a string: ";
    getline(cin, input);

    bool isNewWord = true;
}

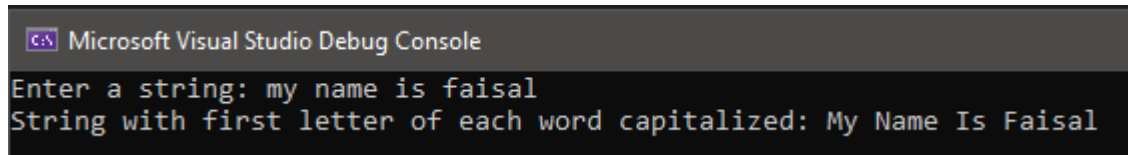
```

```

for (int i = 0; i < input.length(); i++) {
    if (isNewWord && input[i] != ' ') {
        if (input[i] >= 'a' && input[i] <= 'z') {
            input[i] = input[i] - 32;
        }
        isNewWord = false;
    }
    else if (input[i] == ' ') {
        isNewWord = true;
    }
}
cout << "String with first letter of each word capitalized: " << input <<
std::endl;
return 0;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter a string: my name is faisal
String with first letter of each word capitalized: My Name Is Faisal

```

Task 7: Write a C++ program to count all the words in a given string

Code

```

#include <iostream>
#include <string>
using namespace std;

int main()
{
    int words = 0;
    int len;
    string string;

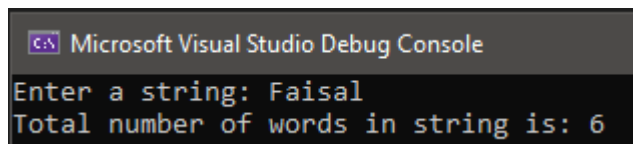
    cout << "Enter a string: ";
    getline(cin, string);

    len = string.length();

    for (int i = 0; i < len; i++) {
        if (string[i] >= 65 && string[i] <= 90 || string[i] >= 97 && string[i] <= 122) {
            words++;
        }
    }
    cout << "Total number of words in string is: " << words << endl;
    return 0;
}

```

Output



```

Microsoft Visual Studio Debug Console
Enter a string: Faisal
Total number of words in string is: 6

```


Lab session 12

Task 1: Create a structure to specify data of an employee. Initialize some random data to employee structure and display that data. Create an array of structures which can store the data of 5 different employees.

Code

```
#include <iostream>
#include <string>
using namespace std;

struct employee {
    string name;
    int id;
    float salary;
    string job_description;
};

int main()
{
    employee e[5];

    cout << "Enter employee details: " << endl;
    for (int i = 0; i < 5; i++) {
        cout << "\nEnter name of employee " << i + 1 << ": ";
        cin >> e[i].name;
        cout << "Enter Id of employee " << i + 1 << ": ";
        cin >> e[i].id;
        cout << "Enter Job Description of employee " << i + 1 << ": ";
        cin >> e[i].job_description;
        cout << "Enter Salary of employee " << i + 1 << ": ";
        cin >> e[i].salary;
    }
    cout << "\nEmployee Details are: " << endl;
    cout << endl;
    for (int i = 0; i < 5; i++) {
        cout << e[i].name << "\t" << e[i].id << "\t" <<
e[i].job_description << "\t" << e[i].salary << endl;
    }
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter employee details:

Enter name of employee 1: Babar
Enter Id of employee 1: 1001
Enter Job Description of employee 1: Teacher
Enter Salary of employee 1: 90000

Enter name of employee 2: Ali
Enter Id of employee 2: 1002
Enter Job Description of employee 2: Teacher
Enter Salary of employee 2: 85000

Enter name of employee 3: Usman
Enter Id of employee 3: 1003
Enter Job Description of employee 3: Teacher
Enter Salary of employee 3: 80000

Enter name of employee 4: Bilal
Enter Id of employee 4: 1004
Enter Job Description of employee 4: Teacher
Enter Salary of employee 4: 95000

Enter name of employee 5: Shan
Enter Id of employee 5: 1005
Enter Job Description of employee 5: Teacher
Enter Salary of employee 5: 90000

Employee Details are:

Babar    1001    Teacher 90000
Ali      1002    Teacher 85000
Usman    1003    Teacher 80000
Bilal    1004    Teacher 95000
Shan     1005    Teacher 90000
```

Task 2: Create a structure to specify data of a student. Initialize some random data to student structure and display that data. Create an array of structures which can store the data of 50 different students.

Code

```
#include <iostream>
#include <string>
using namespace std;

struct student_data{
    string name;
    string roll_num;
    char section;
};

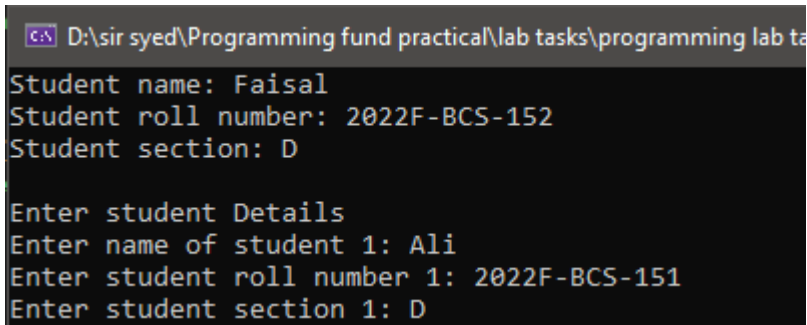
int main() {
    student_data s1 = { "Faisal", "2022F-BCS-152", 'D' };
    cout << "Student name: " << s1.name << endl;
    cout << "Student roll number: " << s1.roll_num << endl;
    cout << "Student section: " << s1.section << endl;
    cout << endl;
    student_data s[50];
    cout << "Enter student Details" << endl;
    for (int i = 0; i < 50; i++) {
        cout << "\nEnter name of student "<<i+1<<": ";
        cin >> s[i].name;
```

```

        cout << "Enter student roll number " << i + 1 << ": ";
        cin >> s[i].roll_num;
        cout << "Enter student section " << i + 1 << ": ";
        cin >> s[i].section;
    }
    cout << "Student details are: " << endl;
    cout << endl;
    for (int i = 0; i < 50; i++) {
        cout << s[i].name << "\t" << s[i].roll_num << "\t" << s[i].section <<
endl;
    }
}

```

Output



```

D:\sir syed\Programming fund practical\lab tasks\programming lab ta
Student name: Faisal
Student roll number: 2022F-BCS-152
Student section: D

Enter student Details
Enter name of student 1: Ali
Enter student roll number 1: 2022F-BCS-151
Enter student section 1: D

```

Task 3: Create a structure to specify data of a cricket player. Initialize some random data to player structure and display that data. Create an array of structures which can store the data of complete squad

Code

```

#include <iostream>
#include <string>
using namespace std;

struct player {
    string name;
    int shirt_num;
    string player_role;
};

int main()
{
    player p[11];

    cout << "Enter Player details: " << endl;
    for (int i = 0; i < 11; i++) {
        cout << "\nEnter name player " << i + 1 << ": ";
        cin >> p[i].name;
        cout << "Enter Shirt number of player " << i + 1 << ": ";
        cin >> p[i].shirt_num;
        cout << "Enter player role " << i + 1 << ": ";
        cin >> p[i].player_role;
    }

    cout << "Player details are: " << endl;
    cout << endl;
    for (int i = 0; i < 11; i++) {
        cout << p[i].name << "\t" << p[i].shirt_num << "\t" <<
p[i].player_role << endl;
    }
}

```

}

}

Output

```
Microsoft Visual Studio Debug Console
Enter Player details:

Enter name player 1: Babar
Enter Shirt number of player 1: 56
Enter player role 1: Batsmen

Enter name player 2: Rizwan
Enter Shirt number of player 2: 15
Enter player role 2: Keeper

Enter name player 3: Haris
Enter Shirt number of player 3: 19
Enter player role 3: Batsmen

Enter name player 4: Shan
Enter Shirt number of player 4: 22
Enter player role 4: Batsmen

Enter name player 5: Nawaz
Enter Shirt number of player 5: 26
Enter player role 5: Allrounder

Enter name player 6: Asif
Enter Shirt number of player 6: 31
Enter player role 6: Batsmen

Enter name player 7: Shadab
Enter Shirt number of player 7: 35
Enter player role 7: Allrounder

Enter name player 8: Naseem
Enter Shirt number of player 8: 40
Enter player role 8: Bowler

Enter name player 9: Shaheen
Enter Shirt number of player 9: 45
Enter player role 9: Bowler

Enter name player 10: Wasim
Enter Shirt number of player 10: 50
Enter player role 10: Bowler

Enter name player 11: Haris
Enter Shirt number of player 11: 60
Enter player role 11: Bowler
Player details are:

Babar    56      Batsmen
Rizwan   15      Keeper
Haris    19      Batsmen
Shan     22      Batsmen
Nawaz    26      Allrounder
Asif     31      Batsmen
Shadab   35      Allrounder
Naseem   40      Bowler
Shaheen  45      Bowler
Wasim    50      Bowler
Haris    60      Bowler
```

Task 4: Create a structure to specify data of customers in a bank. The data to be stored is: Account number, Name, Balance in account. Assume maximum of 10 customers in the bank.

a. Write a function to print the details of all the customers.

b. Write a function to print the Account number and name of each customer with balance below Rs.1000.

Code

```
#include <iostream>
#include <string>
using namespace std;
void print_details(struct Customers s[]);
void low_balance(struct Customers s[]);

struct Customers{
    int acc_num;
    string name;
    int balance;
};

int main() {

    Customers s[10];
    cout << "Enter customer details" << endl;
    for (int i = 0; i < 10; i++) {
        cout << "\nEnter account number of customer "<<i+1<<": ";
        cin >> s[i].acc_num;
        cout << "Enter name of customer " << i + 1 << ": ";
        cin >> s[i].name;
        cout << "Enter balance in account " << i + 1 << ": ";
        cin >> s[i].balance;
    }

    print_details(s);
    low_balance(s);
}

void print_details(struct Customers s[]) {
    cout << "Customer details are: " << endl;
    cout << endl;
    for (int i = 0; i < 10; i++) {
        cout << s[i].acc_num << "\t" << s[i].name << "\t" << s[i].balance << endl;
    }
}

void low_balance(struct Customers s[]) {
    cout << "Customers having low balance: " << endl;
    for (int i = 0; i < 10; i++) {
        if (s[i].balance < 1000) {
            cout << "\nAccount number: " << s[i].acc_num << endl;
            cout << "Name: " << s[i].name << endl;
        }
    }
}
```

Output

```
Microsoft Visual Studio Debug Console
Enter customer details

Enter account number of customer 1: 1001
Enter name of customer 1: Bilal
Enter balance in account 1: 9900

Enter account number of customer 2: 1002
Enter name of customer 2: Rafay
Enter balance in account 2: 5000

Enter account number of customer 3: 1003
Enter name of customer 3: Areeb
Enter balance in account 3: 9800

Enter account number of customer 4: 1004
Enter name of customer 4: Ahsan
Enter balance in account 4: 500

Enter account number of customer 5: 1005
Enter name of customer 5: Jawad
Enter balance in account 5: 900

Enter account number of customer 6: 1006
Enter name of customer 6: Saad
Enter balance in account 6: 9500

Enter account number of customer 7: 1007
Enter name of customer 7: Amin
Enter balance in account 7: 8100

Enter account number of customer 8: 1008
Enter name of customer 8: Usman
Enter balance in account 8: 7700

Enter account number of customer 9: 1009
Enter name of customer 9: Abid
Enter balance in account 9: 5600

Enter account number of customer 10: 1010
Enter name of customer 10: Shan
Enter balance in account 10: 4000
Customer details are:

1001    Bilal    9900
1002    Rafay    5000
1003    Areeb    9800
1004    Ahsan     500
1005    Jawad     900
1006    Saad     9500
1007    Amin     8100
1008    Usman    7700
1009    Abid     5600
1010    Shan     4000
Customers having low balance:

Account number: 1004
Name: Ahsan

Account number: 1005
Name: Jawad
```

Task 5: Write a menu driven program that depicts the working of a library. The menu options should be:

- a) Add book information
- b) Display book information
- c) List all books of given author
- d) List the title of specified book
- e) List the count of books in the library
- f) List the books in the order of book id number
- g) Exit

Create a structure called library to hold book id number, title of the book, authorname, number of pages and price of the book.

Code

```
#include <iostream>
#include <string>
using namespace std;

struct library{
    int id_num;
    string title;
    string author_name;
    int num_pages;
    int price;
};

int main() {
    int choice = 0;
    char y;

    library l[10];
    int num_books = 0;
    cout << "\t\tLibrary Management system" << endl;

    do {

        cout << "\nChoose from the following options: " << endl;
        cout << "1) Add book information" << endl;
        cout << "2) Display book information" << endl;
        cout << "3) List all books of given author" << endl;
        cout << "4) List the title of specified book" << endl;
        cout << "5) List the count of books in the library" << endl;
        cout << "6) List the books in the order of book id number" << endl;
        cout << "7) Exit" << endl;
        cin >> choice;

        switch (choice)
        {
            case 1: {
                cout << "Enter book information" << endl;
                cout << "\nEnter id number of book: ";
                cin >> l[num_books].id_num;
                cout << "Enter title of book: ";
                cin >> l[num_books].title;
                cout << "Enter name of author: ";
                cin >> l[num_books].author_name;
                cout << "Enter number of pages of book: ";
                cin >> l[num_books].num_pages;
                cout << "Enter price of book: ";
                cin >> l[num_books].price;
                cout << endl;
                num_books++;
                cout << "Book added succesfully" << endl;
                break;
            }
            case 2: {
                int id_num;
                cout << "Enter the ID of the book you want to display: ";
                cin >> id_num;
```

```

        cout << endl;
        for (int i = 0; i < num_books; i++) {
            if (id_num == l[i].id_num) {
                cout << "Id number of book: " << l[i].id_num << endl;
                cout << "Title of book: " << l[i].title << endl;
                cout << "Name of author: " << l[i].author_name << endl;
                cout << "Number of pages of book: " << l[i].num_pages << endl;
                cout << "Price of book: " << l[i].price << endl;
            }
        }
        break;
    }
    case 3: {
        string name;
        cout << "Enter name of author: ";
        cin >> name;
        cout << endl;
        for (int i = 0; i < num_books; i++) {
            if (name == l[i].author_name) {
                cout << "\nId number of book: " << l[i].id_num << endl;
                cout << "Title of book: " << l[i].title << endl;
                cout << "Name of author: " << l[i].author_name << endl;
                cout << "Number of pages of book: " << l[i].num_pages << endl;
                cout << "Price of book: " << l[i].price << endl;
            }
        }
        break;
    }
    case 4: {
        int id;
        cout << "Enter the ID of the book you want to display: ";
        cin >> id;
        cout << endl;
        for (int i = 0; i < num_books; i++) {
            if (id == l[i].id_num) {
                cout << "Title of book: " << l[i].title << endl;
            }
        }
        break;
    }
    case 5:
        cout << "Total number of books in library is: " << num_books << endl;
        break;
    case 6: {
        for (int i = 0; i < num_books - 1; i++) {
            for (int j = i + 1; j < num_books; j++) {
                if (l[i].id_num > l[j].id_num) {
                    library temp = l[i];
                    l[i] = l[j];
                    l[j] = temp;
                }
            }
        }
        for (int i = 0; i < num_books; i++) {
            cout << "\nId number of book: " << l[i].id_num << endl;
            cout << "Title of book: " << l[i].title << endl;
            cout << "Name of author: " << l[i].author_name << endl;
            cout << "Number of pages of book: " << l[i].num_pages << endl;
            cout << "Price of book: " << l[i].price << endl;
        }
        break;
    }
    case 7: {
        return 0;
        break;
    }
    default:
        cout << "Wrong choice, Try Again!!" << endl;
}
cout << "\nDo you want to perform any other operation (y/n): ";
cin >> y;

```



```
}  
while (y == 'Y' || y == 'y');
```

```
}
```

Output

```
D:\sir syed\Programming fund practical\lab tasks\programming lab task\64\De  
Library Management system  
Choose from the following options:  
1) Add book information  
2) Display book information  
3) List all books of given author  
4) List the title of specified book  
5) List the count of books in the library  
6) List the books in the order of book id number  
7) Exit  
1  
Enter book information  
Enter id number of book: 1007  
Enter title of book: C++  
Enter name of author: John  
Enter number of pages of book: 5000  
Enter price of book: 3500  
Book added succesfully  
Do you want to perform any other operation (y/n): y  
Choose from the following options:  
1) Add book information  
2) Display book information  
3) List all books of given author  
4) List the title of specified book  
5) List the count of books in the library  
6) List the books in the order of book id number  
7) Exit  
1  
Enter book information  
Enter id number of book: 1005  
Enter title of book: Physics  
Enter name of author: Jason  
Enter number of pages of book: 3000  
Enter price of book: 2700  
Book added succesfully  
Do you want to perform any other operation (y/n): y  
Choose from the following options:  
1) Add book information  
2) Display book information  
3) List all books of given author  
4) List the title of specified book  
5) List the count of books in the library  
6) List the books in the order of book id number  
7) Exit  
2  
Enter the ID of the book you want to display: 1005  
Id number of book: 1005  
Title of book: Physics  
Name of author: Jason  
Number of pages of book: 3000  
Price of book: 2700
```

Do you want to perform any other operation (y/n): y

Choose from the following options:

- 1) Add book information
- 2) Display book information
- 3) List all books of given author
- 4) List the title of specified book
- 5) List the count of books in the library
- 6) List the books in the order of book id number
- 7) Exit

3

Enter name of author: John

Id number of book: 1007

Title of book: C++

Name of author: John

Number of pages of book: 5000

Price of book: 3500

Do you want to perform any other operation (y/n): y

Choose from the following options:

- 1) Add book information
- 2) Display book information
- 3) List all books of given author
- 4) List the title of specified book
- 5) List the count of books in the library
- 6) List the books in the order of book id number
- 7) Exit

4

Enter the ID of the book you want to display: 1005

Title of book: Physics

Do you want to perform any other operation (y/n): y

Choose from the following options:

- 1) Add book information
- 2) Display book information
- 3) List all books of given author
- 4) List the title of specified book
- 5) List the count of books in the library
- 6) List the books in the order of book id number
- 7) Exit

5

Total number of books in library is: 2

Do you want to perform any other operation (y/n): y

Choose from the following options:

- 1) Add book information
- 2) Display book information
- 3) List all books of given author
- 4) List the title of specified book
- 5) List the count of books in the library
- 6) List the books in the order of book id number
- 7) Exit

6

Id number of book: 1005

Title of book: Physics

Name of author: Jason

Number of pages of book: 3000

Price of book: 2700

Id number of book: 1007

Title of book: C++

Name of author: John

Number of pages of book: 5000

Price of book: 3500

Lab session 13

Task 1: What would be the output of the following function, point out the errors, if any.

```
#include <iostream>
using namespace std;
int main()
{
    int firstvalue = 5, secondvalue = 15;
    int* p1, * p2;
    p1 = &firstvalue; // p1 = address of firstvalue
    p2 = &secondvalue; // p2 = address of secondvalue
    *p1 = 10; // value pointed to by p1 = 10
    *p2 = *p1; // value pointed to by p2 = value pointed to by p1
    p1 = p2; // p1 = p2 (value of pointer is copied)
    *p1 = 20; // value pointed to by p1 = 20
    cout << "firstvalue is " << firstvalue << '\n';
    cout << "secondvalue is " << secondvalue << '\n';
    return 0;
}
```

Output

```
firstvalue is 10
secondvalue is 20
```

Task 2: Write a C++ program by using pointer that receives array of 5 integers from keyboard and calculate the sum, average and standard deviation of these numbers.

Code

```
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    int arr[5];
    int* p = arr;
    cout << "Enter 5 digit: ";
    for (int i = 0; i < 5; i++) {
        cin >> arr[i];
    }
    int sum = 0;
    for (int i = 0; i < 5; i++) {
        sum += *(p + i);
    }
    double avg = sum / 5;
    double sd = 0;
    for (int i = 0; i < 5; i++)
    {
        sd += (*(p + i) - avg) * (*(p + i) - avg);
    }
    sd /= 5;
    sd = sqrt(sd);
    cout << "sum is: " << sum << endl;
    cout << "average: " << avg << endl;
    cout << "Standard daviation: " << sd << endl;
}
```

Output

```
Enter 5 digit: 22
32
43
55
64
sum is: 216
average: 43
Standard daviation: 15.146
```

Task 3: Modify the solution of task 1 in order to print the elements of the array in reverse order by using a pointer.

Code

```
#include <iostream>
using namespace std;
int main()
{
    int array[] = { 1, 2, 3, 4, 5 };
    int n = sizeof(array) / sizeof(array[0]);
    int* p1;
    p1 = &array[n - 1];

    for (int i = 0; i < n; i++)
    {
        cout << *p1 << " ";
        p1--;
    }
}
```

Output

```
5 4 3 2 1
```